	Raffles Institution Physics Department Revision Package 2	**Superposition**	**11**

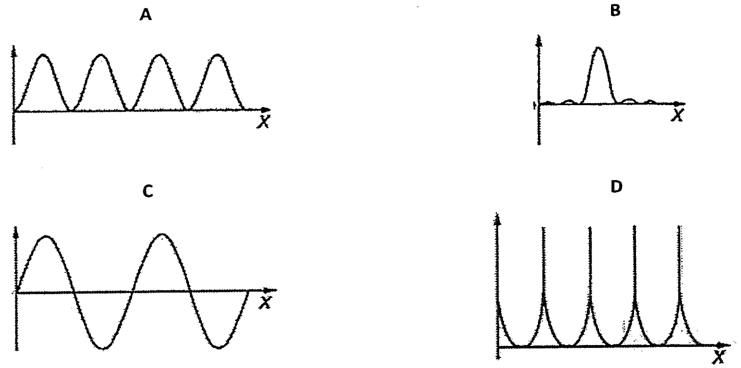
Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs

1. [ACJC/Prelims 2014/P1/25]

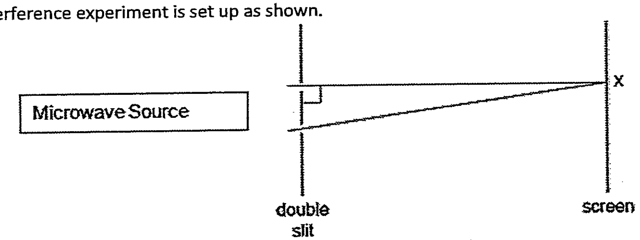
Two coherent monochromatic waves of equal amplitude are brought together to form an interference pattern on a screen.

Which of the following graphs could represent the variation of intensity with position x across the pattern of fringes?



2. [[ACJC/Prelims 2014/P1/26]

A double-slit interference experiment is set up as shown.



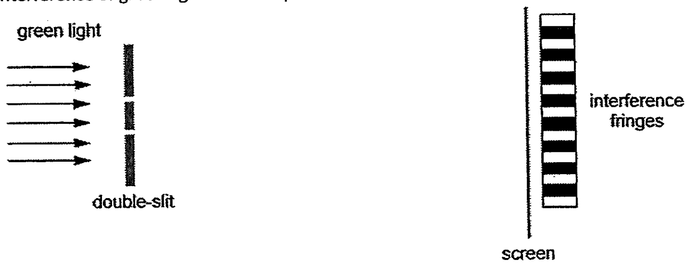
Fringes are formed on the screen which is 40.0 cm away. The distance between the slits is 9.0 cm apart. The microwave source has a wavelength of 2.0 cm and intensity of I.

What is the resultant intensity at X?

A zero **B** I **C** $2I$ **D** $4I$

3. [AJC/Prelims 2014/P1/21]

A double-slit interference of green light was set up as shown and interference fringes are formed on the screen.



Which change would increase the distance between adjacent fringes?

A Use orange light.

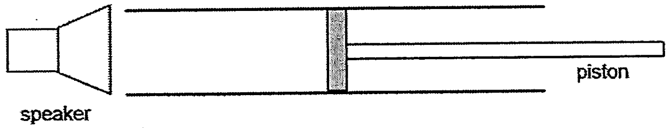
B Reduce the width of each slit.

C Use a double-slit where the slits are further apart.

D Move the double-slit closer to the screen.

4. [AJC/Prelims 2014/P1/22]

A speaker producing sound of frequency 2500 Hz is placed at the open end of a closed pipe containing a gas.



As the piston is moved along the pipe, a series of 6 loud sounds was heard. The first loud sound was observed when the piston was 4.0 cm away from the open end, and the sixth loud sound was observed when the piston was 37 cm away from the open end.

What is the speed of sound in the gas?

A 270 m s^{-1} **B** 330 m s^{-1} **C** 370 m s^{-1} **D** 410 m s^{-1}

5. [CJC/Prelims 2014/P1/19]

A stationary sound wave is set up in a 4.0 m column, with both ends open, using a sound generator producing waves of frequency 400 Hz. The speed of the sound is 320 m s^{-1}.

If the sound generator doubles the frequency of the sound wave produced, how many more antinodes will be present in the stationary wave?

A 5 **B** 10 **C** 11 **D** 21

6. **[CJC/Prelims 2014/P1/21]**

A blue laser light is used in a Young's double-slit experiment.

Which of the following will be observed when a change is made to the experiment?

	change to experiment	observation
A	Covering one of the slits completely.	No fringe pattern is seen.
B	Moving the source of light nearer to the double slits.	Fringe separation will increase.
C	Covering one of the slits with a polaroid.	No change in position of central bright fringe.
D	Replacing the blue laser light with a red laser light.	Central bright fringe will shift upwards.

7. **[CJC/Prelims 2014/P1/22]**

The mercury spectrum contains two intense yellow lines with wavelengths of 577.0 nm and 579.1 nm, respectively. Light from a mercury lamp is incident normally on a diffraction grating ruled with 4300 lines per centimetre. To distinguish the two yellow lines clearly, it is desirable that there should be an angular separation of at least 0.30° between the two lines.

Which order of diffraction can be used?

A 2^{nd} order **B** 3^{rd} order **C** 4^{th} order **D** 5^{th} order

8. **[DHS/Prelims 2014/P1/21]**

The diagram shows a standing wave on a string. The standing wave has three nodes N_1, N_2 and N_3.

Which statement is correct?

A All points on the string vibrate in phase.

B All points on the string vibrate with the same amplitude.

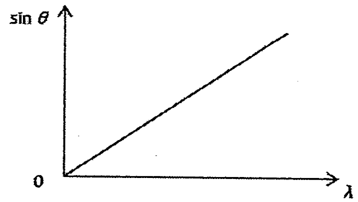
C Points equidistant from N_2 vibrate with the same frequency and in phase.

D Points equidistant from N_2 vibrate with the same frequency and the same amplitude.

9. **[DHS/Prelims 2014/P1/22]**

A diffraction grating with N lines per metre is used to deflect light of various wavelengths λ.

The diagram shows a relation between the deflection angles of θ for different values of λ in the n^{th} order interference pattern.

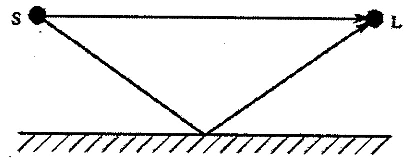


What is the gradient of the graph?

A Nn **B** $\frac{N}{n}$ **C** $\frac{n}{N}$ **D** $\frac{1}{Nn}$

10. **[HCI/Prelims 2014/P1/17]**

A loudspeaker at position S emits sound of a single frequency. The sound travels to Leo who is at position L, both through a straight path and after reflection from a wall as shown.



As Leo walks directly towards the wall, the sound alternates between loud and soft.

Which change would result in an increase in the distance between loud and soft sounds?

A Increasing the frequency emitted by the loudspeaker.

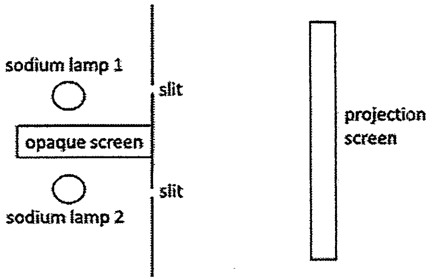
B Moving the loudspeaker closer to the wall.

C Moving the loudspeaker towards L.

D Increasing the loudness of the sound emitted by the loudspeaker.

11. [HCI/Prelims 2014/P1/18]

A sodium lamp produces two distinct yellow lines in the visible part of the spectrum, at 589.0 nm and 589.6 nm. Two such lamps are separated by a screen. The light of both lamps pass through a slit each, as indicated in the figure. It turns out that no interference pattern can be obtained on the projection screen.



Which of the following explains the observation?

A The lamps are not point sources.

B The light from the lamps do not have exactly the same amplitude.

C The light from the lamps is not coherent.

D The light from the lamps is not monochromatic.

12. [HCI/Prelims 2014/P1/19]

Two samples of guitar strings, one 60.0 cm long and the other 30.0 cm long are plucked in the presence of a tuning fork of frequency 256 Hz. The wave velocity v is 153.5 m s^{-1} in both strings.

Which of the following is true?

A Only the 30.0 cm string causes the tuning fork to vibrate.

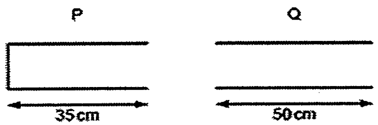
B Only the 60.0 cm string causes the tuning fork to vibrate.

C Both strings cause the tuning fork to vibrate.

D Neither string cause the tuning fork to vibrate.

13. [IJC/Prelims 2014/P1/21]

Travelling waves of wavelength 20 cm are created in the air columns in a closed pipe P and an open pipe Q. The lengths of the pipes are shown.

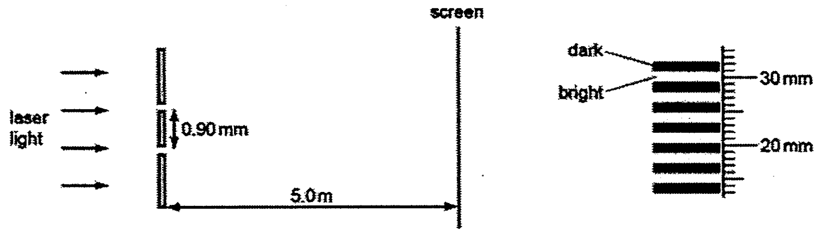


In which pipe or pipes are stationary waves formed?

A P and Q B P only C Q only D neither P nor Q

14. [IJC/Prelims 2014/P1/22]

The diagrams show the arrangement of the apparatus in a Young's double slit experiment and also part of the pattern formed on the screen with a ruler placed next to it.

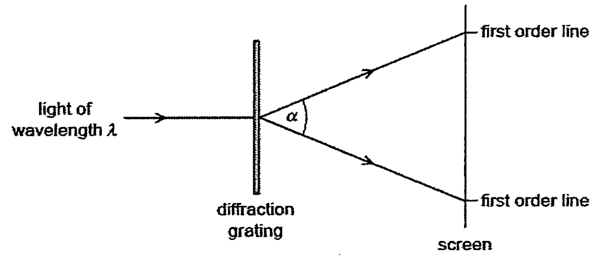


What is the wavelength of the light?

A 4.8×10^{-7} m B 5.4×10^{-7} m C 3.2×10^{-6} m D 3.4×10^{-6} m

15. [IJC/Prelims 2014/P1/23]

Light of wavelength λ passes through a diffraction grating with slit spacing d. A series of lines is observed on a screen.



What is the angle α between the two first order lines?

A $\sin^{-1}\left(\frac{\lambda}{2d}\right)$ B $\sin^{-1}\left(\frac{\lambda}{d}\right)$ C $2\sin^{-1}\left(\frac{\lambda}{2d}\right)$ D $2\sin^{-1}\left(\frac{\lambda}{d}\right)$

16. [IJC/Prelims 2014/P1/26]

A wave is diffracted as it passes through an opening in a barrier.

The amount of diffraction that the wave undergoes depends on both the

A amplitude and frequency of the incident wave.

B wavelength and speed of the incident wave.

C wavelength of the incident wave and the size of the opening.

D amplitude of the incident wave and the size of the opening.

17. [JJC/Prelims 2014/P1/27]

Which of the following is always true for a stationary wave?

A All particles have the same amplitude.

B No energy is transferred from one end of the wave to the other.

C The particles in the wave oscillate perpendicularly to the direction of wave travel.

D All particles vibrate in simple harmonic motion with the same frequency as the wave.

18. [MJC/Prelims 2014/P1/19]

A sound source is placed several metres from a plane reflecting wall in a large chamber containing a gas. A microphone connected to a cathode-ray oscilloscope (CRO) is used to detect nodes and antinodes along the line from the source to the wall. The microphone is moved from one node through 15 antinodes to another node, a distance of 1.2 m.

On the CRO, the time-base is set to 0.50 ms cm^{-1}, and the distance between 5 consecutive crests is 8.5 cm.

What is the speed of sound in this gas?

A 150 m s^{-1} **B** 300 m s^{-1} **C** 320 m s^{-1} **D** 400 m s^{-1}

19. [MJC/Prelims 2014/P1/20]

A narrow beam of monochromatic light falls at normal incidence on a diffraction grating. Second-order diffracted beams are formed at angles of 32° to the original direction.

How many bright spots can be seen on the screen?

A 3 **B** 4 **C** 7 **D** 9

20. [NJC/Prelims 2014/P1/21]

Plane sound waves of frequency 100 Hz fall normally on a smooth wall.

At what distances from the wall will the air particles have maximum amplitude of vibration. Assume the speed of sound to be 340 m s^{-1}.

A 0.85 m, 2.55 m, 4.25 m, ...

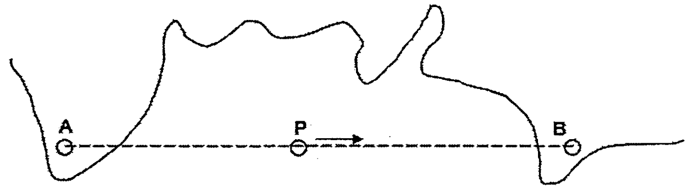
B 1.70 m, 3.40 m, 5.10 m, ...

C 2.00 m, 4.00 m, 6.00 m, ...

D 2.50 m, 5.00 m, 7.50 m, ...

21. [NJC/Prelims 2014/P1/22]

The diagram shows a coastline with two radio transmitting stations **A** and **B**. The stations are 100 km apart and each behaves as a point source of E.M. waves which radiate as circular wavefronts. Both stations transmit at a frequency of 1.5 MHz and have identical powers.



Ship **P** travels a straight line course between the stations **A** and **B** at a steady speed of 2.0 m s^{-1}. The combined signal received by the ship rises and falls at regular intervals. As the ship passes the point midway between **A** and **B** the amplitude of the combined signal received from both stations is a minimum.

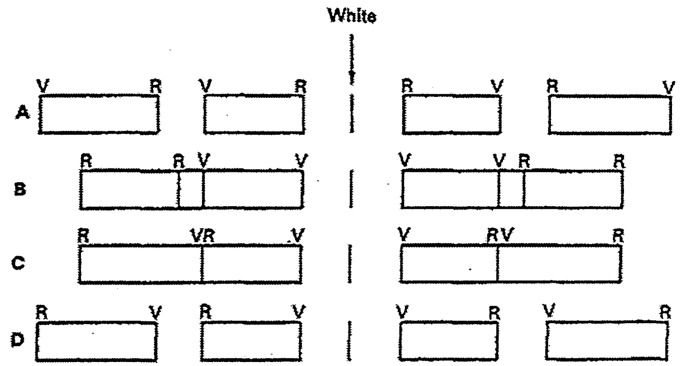
What is the frequency of the received signal to pass from one minimum to the next?

A 0.01 Hz **B** 0.02 Hz **C** 50 Hz **D** 100 Hz

22. [NJC/Prelims 2014/P1/23]

A narrow shaft of white light (from violet 400 nm (V), to red 700 nm (R)) falls with normal incidence on a transmission grating, and produces two orders of spectra on a distant screen.

Which diagram shows the appearance of the spectra on the screen?



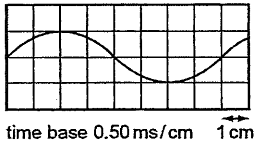
23. [NYJC/Prelims 2014/P1/20]

A stationary sound wave is set up between a loudspeaker and a wall. A microphone is connected to a cathode-ray oscilloscope (c.r.o.) and is moved along a line directly between the loudspeaker and the wall.

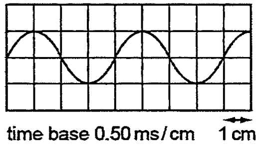
The amplitude of the trace on the c.r.o. rises to a maximum at a position X, falls to a minimum and then rises once again to a maximum at a position Y. The distance between X and Y is 33 cm. The speed of sound in air is 330 m s^{-1}.

Which diagram represents the c.r.o. trace of the sound received at X?

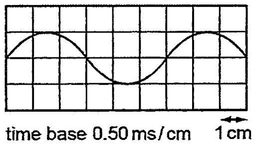
A



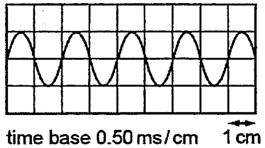
B



C

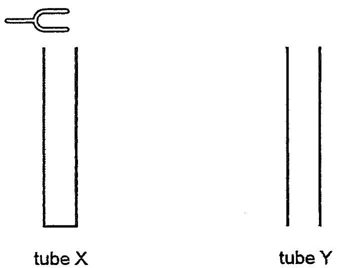


D



24. [NYJC/Prelims 2014/P1/21]

The diagram shows two tubes. The tubes are identical except tube X is closed at its lower end while tube Y is open at its lower end. Both tubes have open upper ends.



A tuning fork placed above tube X causes resonance of the air at frequency f. No resonance is found at any lower frequency than f with tube X.

Which tuning fork will produce resonance when placed just above tube Y?

A a fork of frequency $f/2$.

B a fork of frequency $2f/3$.

C a fork of frequency $3f/2$.

D a fork of frequency $2f$.

25. [NYJC/Prelims 2014/P1/22]

When the light from two lamps falls on a screen, no interference pattern can be obtained.

This is because the

A lamps are not point sources.

B lamps emit light of different amplitudes.

C light from the lamps is not coherent.

D light from the lamps is white.

26. [PJC/Prelims 2014/P1/20]

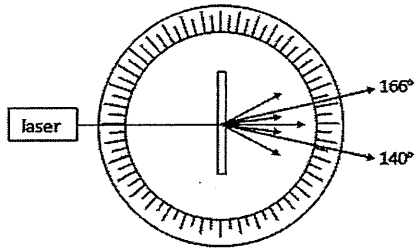
In a two-slit interference experiment, one slit transmits waves of twice the amplitude compared to the other slit.

If the maximum intensity of the interference pattern is I_o, what is the minimum intensity of the pattern?

A zero **B** $\frac{I_o}{2}$ **C** $\frac{I_o}{4}$ **D** $\frac{I_o}{9}$

27. [PJC/Prelims 2014/P1/217]

Light from a laser is directed normally at a diffraction grating as shown in the figure below. The diffraction grating is situated at the centre of the circular scale, marked in degrees. The readings on the scale for the second order diffracted beams are 140° and 166°. The wavelength of the laser light is 500 nm.

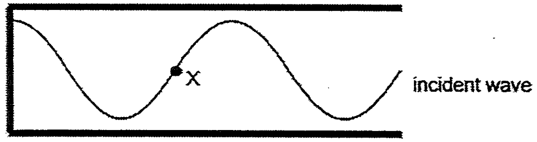


What is the spacing of the slits of the diffraction grating?

A 1.14×10^{-6} m **B** 2.22×10^{-6} m **C** 2.28×10^{-6} m **D** 4.45×10^{-6} m

28. **[RVHS/Prelims 2014/P1/20]**

A continuous progressive sound wave is incident on the air column with a closed end. The diagram shows just the incident wave at one instant.

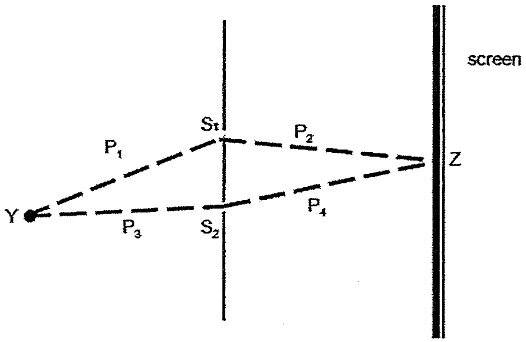


Which pair correctly describes the amplitudes of displacement and pressure change at point X where the stationary sound wave forms?

	displacement amplitude	pressure change amplitude
A	maximum	maximum
B	minimum	maximum
C	maximum	minimum
D	minimum	minimum

29. **[RVHS/Prelims 2014/P1/19]**

Two identical narrow slits S_1 and S_2 are illuminated by light from a point source Y.



If the light of wavelength λ is allowed to fall on a screen as shown in the diagram above, and n is a positive integer, the condition for destructive interference at Z is

A $(P_2 - P_4) = (2n + 1)\ \lambda/2$

B $(P_3 + P_4) - (P_2 + P_1) = (2n + 1)\ \lambda/2$

C $(P_4 - P_2) + (P_1 - P_3) = (2n + 1)\ \lambda/2$

D $(P_3 + P_4) - (P_2 + P_1) = n\ \lambda/2$

30. **[RVHS/Prelims 2014/P1/21]**

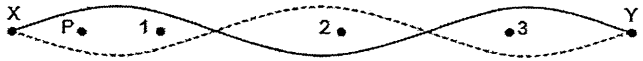
Microwaves of wavelength 3.30 cm are incident normally on a row of parallel metal rods. The distance between the rods and the microwave detector is 33.0 cm. The first order diffraction maxima is observed at an angle of 7.4° to the direction of the incident waves.

What is the angle between the third order diffraction maxima?

A 14.9° **B** 22.2° **C** 22.7° **D** 45.4°

31. **[SAJC/Prelims 2014/P1/21]**

A standing wave is set up on a stretched string XY as shown in the diagram.

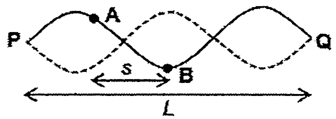


At which point(s) will the oscillation be exactly in phase with that at point P?

A 1 and 3 only **B** 2 and 3 only **C** 1 only **D** 2 only

32. **[SAJC/Prelims 2014/P1/22]**

A guitar string of length L is stretched between two fixed points **P** and **Q** and made to vibrate transversely as shown.



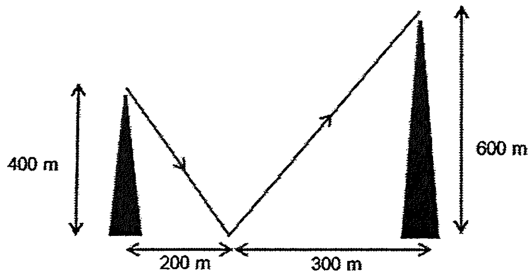
Two particles **A** and **B** on the string are separated by a distance s. The maximum kinetic energies of **A** and **B** are K_A and K_B respectively.

Which row gives the correct phase difference and maximum kinetic energies of the particles?

	phase difference	maximum kinetic energy
A	$\left(\frac{3s}{2L}\right)\times 360°$	same
B	$\left(\frac{3s}{2L}\right)\times 360°$	$K_A < K_B$
C	180°	same
D	180°	$K_A < K_B$

33. [SRJC/Prelims 2014/P1/21]

The figure below shows a radio wave transmitter at a height of 400 m above ground and a receiver 600 m above the ground. The transmitter and receiver are at a distance of 500 m apart. The receiver can receive signals directly from the transmitter, and indirectly from signals that bounce off the ground as shown in the diagram. The ground is level between the transmitter and receiver and the wave undergoes a 180° phase change upon reflection.

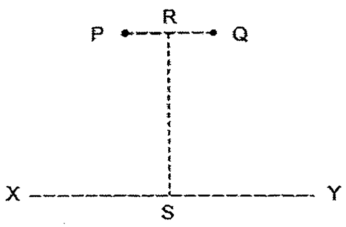


What is the longest wavelength that can cause constructive interference at the receiver?

A 580 m B 618 m C 1160 m D 1240 m

34. [SRJC/Prelims 2014/P1/22]

Coherent waves of equal amplitude are produced at P and Q and travel outwards in all directions. The line RS is halfway between P and Q and perpendicular to the line joining P and Q. The distance RS is much greater than the distance PQ.

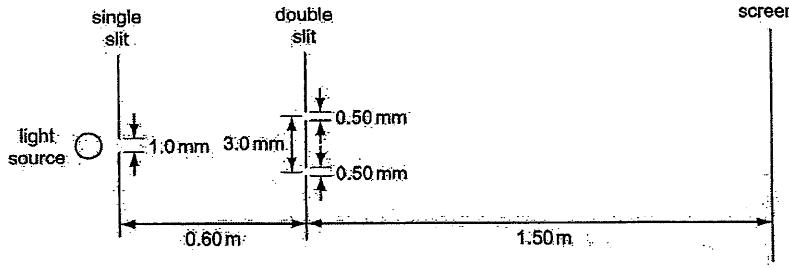


Along which line can we observe (i) a stationary wave (ii) an interference pattern (iii) an anti-nodal line?

	stationary wave	interference pattern	anti-nodal line
A	PQ	RS	XY
B	XY	RS	PQ
C	RS	XY	PQ
D	PQ	XY	RS

35. [TJC/Prelims 2014/P1/23]

A student sets up an experiment to demonstrate double slit interference, using light of wavelength 6.0×10^{-7} m. The main features of the apparatus, and some of the dimensions, are shown in the diagram below.

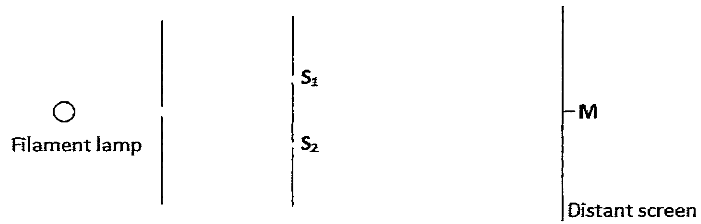


What is the separation of the bright fringes on the screen?

A 0.12 mm B 0.30 mm C 0.90 mm D 1.80 mm

36. [TPJC/Prelims 2014/P1/22]

The diagram shows a Young's double slits experiment. M is at the centre of the distant screen where a bright fringe occurs. The two slits S_1 and S_2 are identical.



(Not to scale)

Which of the following is the best estimate of the ratio of the intensity at M when both slits are in use, to the intensity at M when S_2 is covered?

A 0.5 B 1 C 2 D 4

37. [TPJC/Prelims 2014/P1/23]

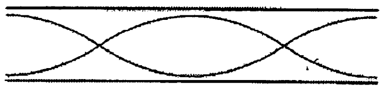
A standing wave is set up on a stretched string **XY** as shown in the diagram below.

At which point(s) will the oscillation be exactly out of phase by π rad with that at point **P**?

A 1, 2 and 3 B 1 and 2 only C 1 only D 2 only

38. [VJC/Prelims 2014/P1/21]

A stationary wave is formed in the air in an open tube as represented in the diagram.

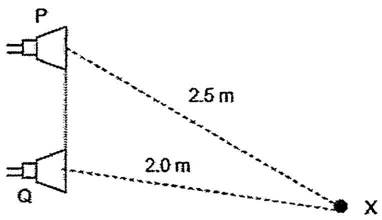


How many antinodes are formed by a stationary wave of twice the frequency?

A 2 B 4 C 5 D 6

39. [VJC/Prelims 2014/P1/22]

Microwaves of wavelength 4.0 cm are produced by two microwave transmitters P and Q operating in phase. Point X is 2.5 m from transmitter P and 2.0 m from transmitter Q as shown in the figure. Microwaves from transmitter P arrives at point X with intensity I and amplitude of oscillation A while the microwaves from transmitter Q arrives at point X with intensity $4I$.



What is the resultant intensity at point X in terms of I?

A zero B I C $3I$ D $9I$

40. [YJC/Prelims 2014/P1/21]

When a two-slit arrangement was set up to produce interference fringes on a screen using a monochromatic source of green light, the fringes were found to be too close together for convenient observation.

In which of the following ways would it be possible to increase the fringe separations?

A Decrease the distance between the slits and the screen

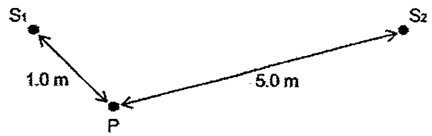
B Increase the separation between the slits

C Increase the width of each slit

D Replace the light source with a monochromatic red light

41. [VJC/Prelims 2014/P1/22]

Water waves of wavelength 0.50 m are produced by two sources S_1 and S_2.



Point P is 1.0 m from S_1 and 5.0 m from S_2. S_1 alone and S_2 alone each produces a wave of amplitude a and $3a$ at P respectively.

Which one of the following is the amplitude of the resultant wave at point P when S_1 and S_2 operates in anti-phase?

A $4a$ B $2a$ C a D zero

II. Structured Questions

1. [ACJC/Prelims 2014/P2/4]

a. Explain what is meant by the term interference. [2]

b. A Kundt's tube is an experimental acoustical instrument that serves to measure the speed of sound in different medium.

It comprises of a long horizontal tube, containing a fine powder, which is closed at one end. A loudspeaker connected to a signal generator is positioned at the other end. The device is shown in Fig 1.1 and the signal generator is set to a frequency of 400 Hz.

From the interference of waves resulting in stationary waves being formed, an interesting pattern can be observed as seen in Fig 1.1.

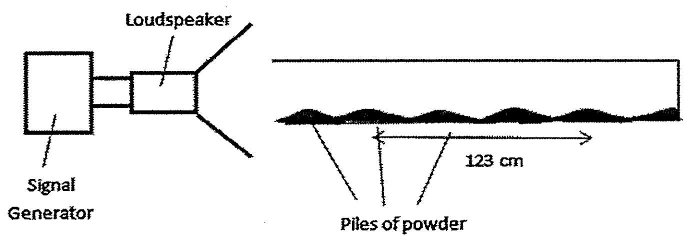


Fig. 1.1

i. Label the positions of nodes (N) and antinodes (A) on Fig. 1.1. [1]

ii. Explain the formation of the piles of fine powder at the positions shown on Fig 1.1. [3]

iii. Hence or otherwise, determine the speed of the sound waves in that medium. [2]

[Total: 8]

2. **[ACJC/Prelims 2014/P3/5]**

A double slit with slit separation d = 0.400 mm is situated a distance 3.2 m from a detector moved vertically along XY as shown in Fig. 2.1. The double slit is illuminated with coherent light of wavelength λ.

a.

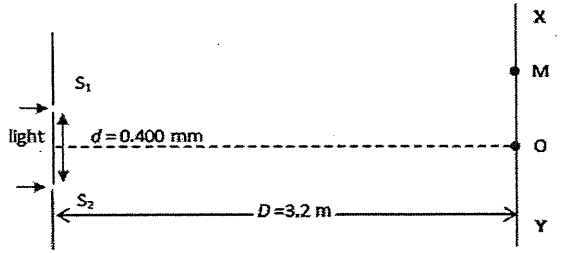


Fig. 2.1

A **detector** was moved along **XY** and the detected intensity is shown in Fig. 2.2.

The relative positions of the peaks are with respect to O, the central maximum.

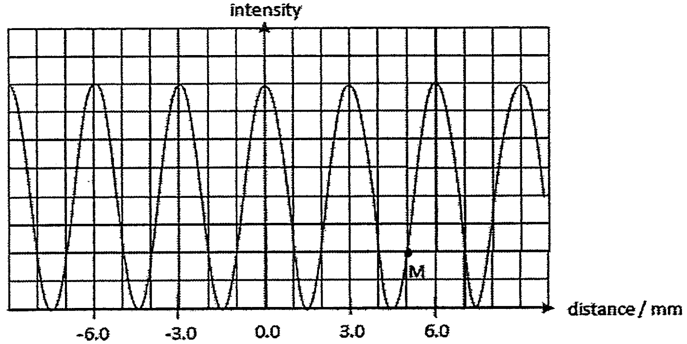


Fig. 2.2

i. Deduce the wavelength λ of the light. [2]

ii. With reference to Fig 2.1, calculate the phase difference between the two waves at point M. [2]

b. Both the slits are covered with two identical thin transparent boxes. The one in front of S_2 is filled with vacuum whereas the one in front of S_1 is filled with gas of refractive index n as shown in Fig. 2.3.

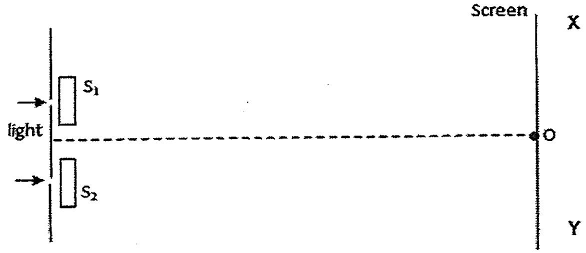


Fig. 2.3

It is considered that the actual path length in the gas to be the product of its refractive index and actual distance travelled d.

The new light intensity pattern along **XY** is shown in Fig. 2.4.

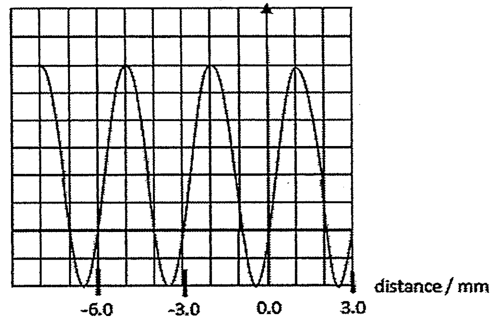


Fig. 2.4

With reference to Fig 2.2 and Fig. 2.4, the central peak has been shifted by 1.0 mm towards M.

By deducing the phase difference at the central position or otherwise, calculate the refractive index, n of the gas if the gas is 1.0×10^{-6} m thick. [2]

[Total: 6]

3. **[AJC/Prelims 2014/P2/3]***

a. Explain what is meant by the principle of superposition. [1]

b. Light from a mercury vapour lamp is incident normally on a diffraction grating and the diffraction pattern is observed on a screen placed 30 cm away from the diffraction grating. The positions of the first order orange spectrum of wavelength 580 nm are 20 cm apart, as shown in Fig. 3.1 below.

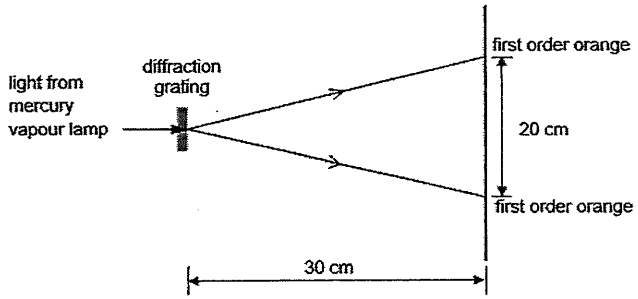


Fig. 3.1

i. Show that the diffraction grating has 545 number of lines per mm. [2]

ii. Fig. 3.2 shows some of the energy levels in a mercury atom. Another spectrum that was also emitted from the mercury lamp is shown by the transition A.

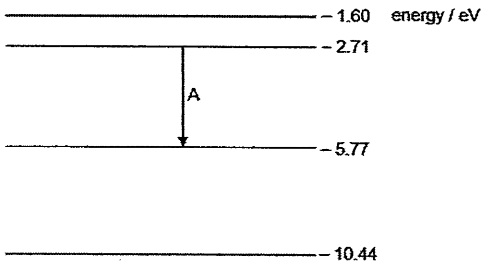


Fig. 3.2

Calculate the wavelength of the spectrum associated with the transition A **[2]**

iii. Hence, determine the highest order of diffracted beam that can be observed on the screen in Fig. 3.1 for the spectrum calculated in **b. ii.**. **[2]**

[Total: 7]

4. [CJC/Prelims 2014/P2/2 Part]

b. A stereo system in a large hall has two identical speakers, S_1 and S_2, 1.2 m apart. The amplitude of the output of each speaker is proportional to the voltage across its terminals. The voltage input to each speaker is adjusted by means of a balance control. The arrangement is shown in Fig. 4.1.

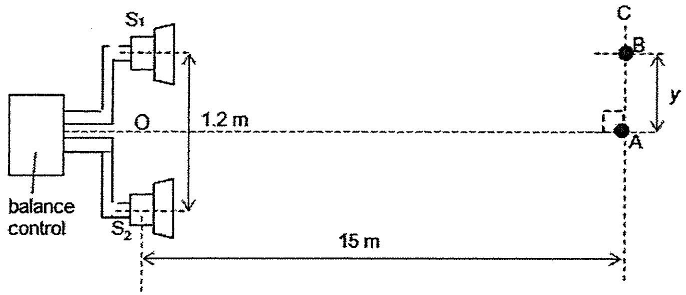


Fig. 4.1 (not to scale)

Initially, the speakers are emitting signals of frequency 1000 Hz which are in phase. The balance control is set such that there is a voltage of 6 V r.m.s. across each speaker. An observer hears a loud sound of intensity I_{max} at point A. As he moves along the line AC, 15 m away from the speakers, he observes that the intensity first falls to zero at point B, a distance y from point A. The speed of sound in air is 330 m s^{-1}.

i. Determine the distance y. **[2]**

ii. Determine the next higher frequency of the speaker such that point B would also be a point of zero intensity. **[3]**

iii. With the speakers emitting the original signal frequency of 1000 Hz, the balance control is now adjusted such that the voltages across S_1 and S_2 are 3.0 V r.m.s. and 9.0 V r.m.s. respectively.

In terms of I_{max}, determine the new intensity at point B. **[3]**

[Total: 8]

5. [DHS/Prelims 2014/P2/2]

a. State two necessary conditions that must be satisfied in order that coherent waves may interfere. **[2]**

b. Light from a light source is incident on a double slit after passing through a single slit, as shown in Fig. 5.1. Interference fringes are observed on the screen.

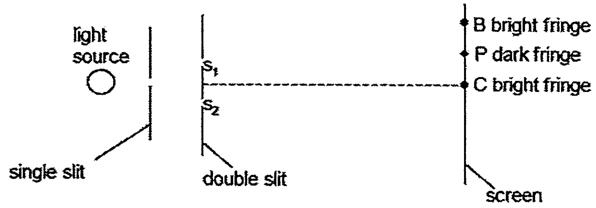


Fig. 5.1 (not to scale)

i. Explain why a single slit is used in the apparatus. **[1]**

ii. The intensity at B due to each wave is I.

Determine, in terms of I, the resultant intensity of the waves at B. **[2]**

iii. State the phase difference between the waves that meet at P. **[1]**

iv. A third slit S_3 is now made at an equal distance below S_2, as shown in Fig. 5.2.

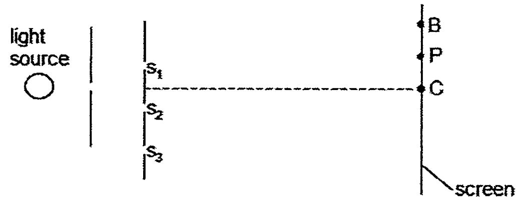


Fig. 5.2 (not to scale)

Explain whether point P on the screen remains dark. **[2]**

[Total: 8]

6. **[HCI/Prelims 2014/P2/3]**

a. While Alvin drives through a long underground tunnel in Singapore, it is obvious that nothing can be seen a few metres into the tunnel without additional sources of light. However, his radio continues playing songs from Gold 90.5FM, which is broadcast at a frequency of 90.5 MHz, and his Global Positioning System (GPS) continues to display his car position. Further into the tunnel, the GPS reports that it has lost its connection with the satellites but his radio continues to play.

i. State the type of electromagnetic waves that Alvin's GPS system uses to communicate with its satellites. Explain how you come to this conclusion. [2]

ii. One particular day, the traffic was particularly heavy in the tunnel. When driving very slowly, Alvin finds that there were many points in the tunnel where the radio connection appears to be lost. Alvin decided to count those points and found there were exactly 400 points.

Assuming that the tunnel is perfectly straight, calculate the distance between the first and the last point where the radio connection is lost. [2]

b. The apparatus illustrated in Fig. 6.1 is used to demonstrate two-source interference using light.

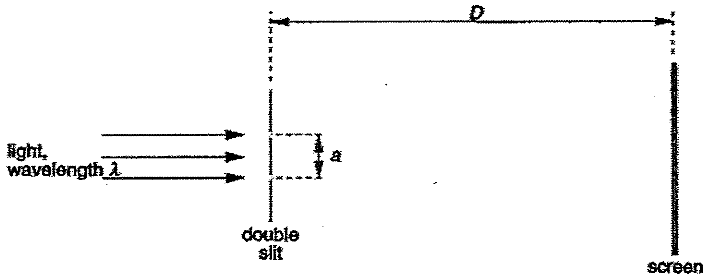


Fig. 6.1 (not to scale)

For a particular experiment, red light of wavelength = 690 nm is used, with a = 0.333 mm and D = 1.20 m.

i. State two conditions that must be satisfied in order that two waves may interfere. [2]

ii. Calculate the fringe separation. [2]

iii. The double slit in the experiment above is now replaced by a diffraction grating, which is labelled 300 lines mm^{-1}.

Determine the highest order of light that is still visible on the screen, given that the screen is flat and 2.40 m wide. [3]

[Total: 11]

7. **[IJC/Prelims 2014/P3/4]**

a. Explain the term interference. [1]

b. A ripple tank is used to demonstrate interference between water waves.

Describe the apparatus used to produce two sources of coherent waves that have circular wavefronts and how the pattern of interfering waves may be observed. [3]

c. Sound waves travel from a source S to a point X along two paths SX and SPX, as shown in Fig. 7.1. The waves undergo a phase change of π radians as it reflects off the surface at P.

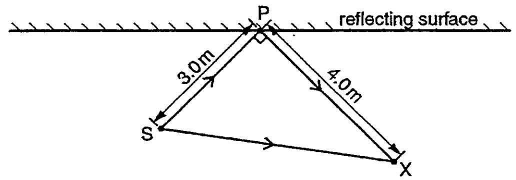


Fig. 7.1

i. The frequency of the sound from S is 400 Hz and the speed of sound is 320 m s^{-1}.

Calculate the wavelength of the sound waves. [2]

ii. The distance SP is 3.0 m and the distance PX is 4.0 m. The angle SPX is 90°.

Suggest whether a maximum or a minimum is detected at point X. Explain your answer. [2]

[Total: 8]

8. **[JJC/Prelims 2014/P3/8 Part]**

b. A wave of amplitude A and intensity I is coincident with a second wave of amplitude 3A. Both waves have the same frequency.

Calculate, in terms of I, the resultant intensity when the phase difference is π rad. [2]

c. To determine the wavelength of sound in air, the sound from a loudspeaker is directed normally towards a reflector. A microphone attached to a cathode-ray-oscilloscope (CRO) is moved along a line between the loudspeaker and the reflector.

i. Using the above set-up, describe briefly a simple procedure to determine the wavelength of sound in air. [2]

ii. Explain why the position of the microphone where the CRO indicates maximum amplitude is considered a displacement node of the air molecules. [1]

d. i. State what is meant by the term coherence of two waves. [1]

ii. Explain how the coherence condition for interference is obtained when a plane wave front meets a pair of narrow slits which are parallel to the wave front. [1]

e. Light of wavelength 5.24 × 10–7 m is shone on a double slit whose separation is 0.220 mm.

Calculate the separation of the fringes on a screen placed 0.945 m from the double slit. The screen and double slit are both placed at right-angles to the beam of light. [2]

f. A diffraction grating with a grating spacing of 2.20×10^{-6} m is used to examine the visible light from a lamp. It is observed that the first order violet light emerges at an angle of 11.8°.

i. Show that the wavelength of the violet light is 4.50×10^{-7} m. [1]

ii. It is also observed that first order red light of wavelength 5.99×10^{-7} m emerges at an angle of 15.8° from the same grating.
Describe and explain what will be visible at an angle of 54.9°. [3]

[Total: 13]

9. **[MJC/Prelims 2014/P3/6 Part]**

a. i. Explain what is meant by

1. diffraction [1]
2. interference [1]
3. coherence [1]

ii. State two conditions for observable interference of two waves. [2]

b. To help guide large ships berth properly into docks, an engineer proposed using interference of electromagnetic (EM) waves. The proposal suggests installing two EM wave emitters P and Q positioned 95 m apart at the edges of the dock gates as shown in Fig. 9.1. The two emitters can be taken to be point sources and they emit EM waves of frequency f_1 in phase.

The ship can be guided by searching for the strong signal radiated along the lines of constructive interference, also known as anti-nodal lines. For safety, it is important for the ship to ensure that it is sailing along the centre-line of the gates, as such the ship needs to "lock on" to the central anti-nodal line.

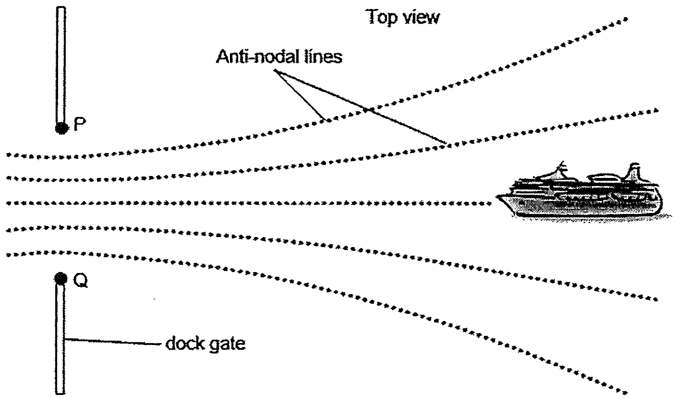


Fig. 9.1 (not to scale)

i. Explain why the centre-line will always be an anti-nodal line regardless of the frequency of the EM waves used. [1]

ii. State and explain which type of EM waves is suitable for such a system. [2]

iii. Assuming that the ship is sailing along the centre-line, state and explain how the intensity of the resultant signal varies as it approaches the dock gates. [2]

c. One particular large cargo ship strays off the centre-line as shown in Fig. 9.2.

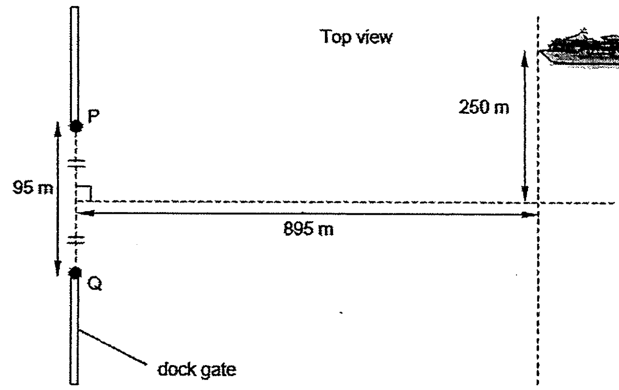


Fig. 9.2 (not to scale)

Explain quantitatively whether the ship is on an anti-nodal line, given that f1 is 23.5 MHz. [3]

d. As an additional precaution to ensure that the ship "locks on" to the central anti-nodal line, the emitters can simultaneously emit another EM wave of a different frequency f_2.

i. Explain how this precaution helps to prevent the ship from "locking on" to the wrong anti-nodal line. [1]

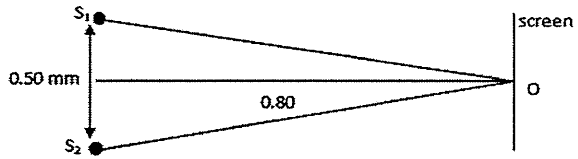
ii. Discuss why this additional precaution may still not be fool proof. [1]

[Total: 15]

10. **[NJC/Prelims 2014/P3/7 Part]**

d. i. Explain the meaning of interference between two separate wave trains. [2]

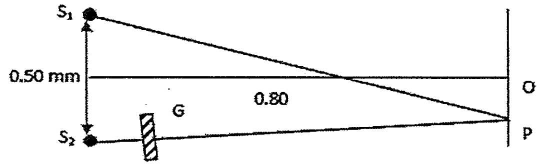
ii. In the figure below, S_1 and S_2 are two coherent light sources in a Young's two slit experiment separated by a distance 0.50 mm and O is a point equidistant from S_1 and S_2. O is on a screen which is 0.80 m from the slits. The wavelength of the monochromatic light from the two slits is 6.0×10^{-7} m.



1. Calculate the fringe separation of the fringes observed. [1]

2. When a thin parallel-sided piece of glass G is placed near S_2 as shown, the centre of the fringe system moves from O to P.

Explain why the centre of the fringe system is shifted. [2]



3. Glass **G** has a thickness of 3.6×10^{-6} m. The refractive index of the glass is 1.5. Calculate the wavelength of light in glass.

Refractive index of glass $= \dfrac{\text{wavelength of light in air}}{\text{wavelength of light in glass}}$ [1]

4. Calculate the increase in the number of waves in G when it replaces air of the same thickness. [2]

5. Using your answers obtained in **d. ii. 1.** and **d. ii. 4.**, determine the length of OP. [2]

[Total: 10]

11. [NYJC/Prelims 2014/P3/7]

a. **i.** State the principle of superposition. [1]

ii. Explain what is meant when two sources are coherent. [1]

b. Two sources S_1 and S_2 of sound are situated 80 cm apart in air, as shown in Fig. 11.1.

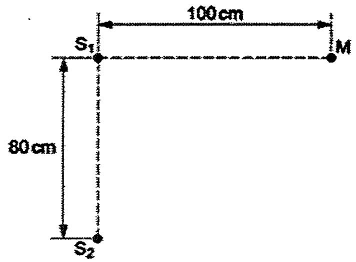


Fig. 11.1

The frequency of vibration can be varied. The two sources always vibrate in phase but have different amplitudes of vibration.

A microphone M is situated a distance 100 cm from S_1 along a line that is normal to S_1S_2.

As the frequency of S_1 and S_2 is gradually increased, the microphone M detects maxima and minima of intensity of sound.

i. State the two conditions that must be satisfied for the intensity of sound at M to be zero. [2]

ii. The speed of sound in air is 330 m s^{-1}. The frequency of sound from S1 and S2 is increased.

Determine the number of minima that will be detected at M as the frequency is increased from 1.0 kHz to 4.0 kHz. [4]

iii. The variation with time of the displacement x of the sound waves arriving at M from S_1 and S_2 are as shown in Fig. 11.2a and Fig. 11.2b respectively.

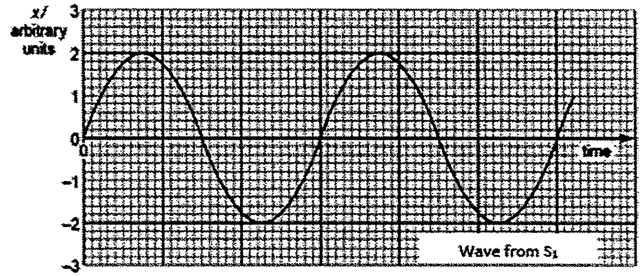


Fig. 11.2a

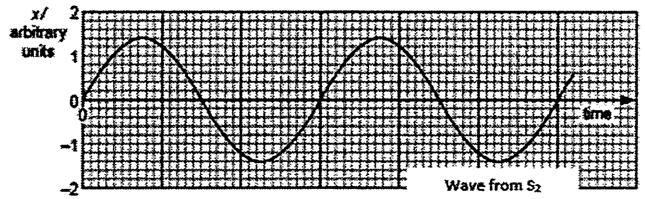


Fig. 11.2b

Determine the ratio of $\dfrac{\text{intensity of sound when a maxima is detected at M}}{\text{intensity of sound when a minima is detected at M}}$ [3]

c. Laser beam of red light of wavelength 644 nm is incident normally on a diffraction grating having 550 lines per millimetre, as illustrated in Fig. 11.3.

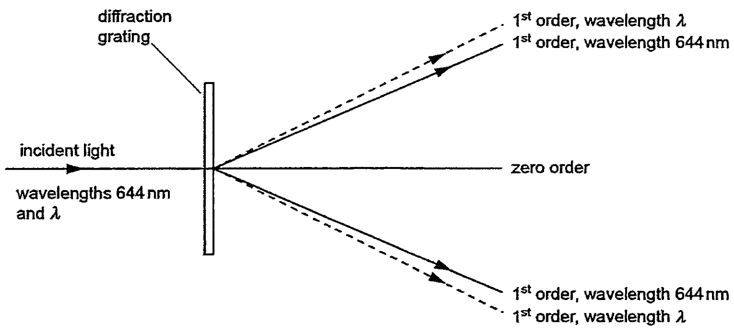


Fig. 11.3

Red light of wavelength λ is also incident normally on the grating. The first order diffracted light of both wavelengths is illustrated in Fig. 11.3.

i. Determine the total number of bright spots of wavelength 644 nm that are visible. [2]

ii. State and explain

1. whether λ is greater or smaller than 644 nm [2]

2. in which order of diffracted light there is the greatest separation of the two wavelengths [2]

iii. The diffraction grating is now rotated 90° about an axis parallel to the incident laser beam, as shown in Fig. 11.4.

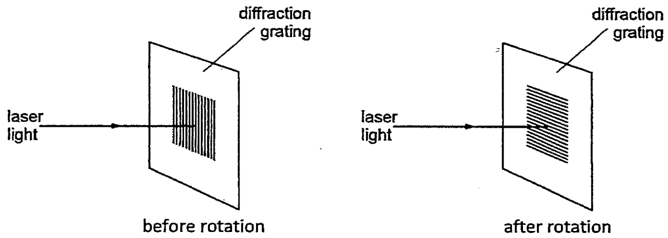


Fig. 11.4

State what effect, if any, this rotation will have on the diffraction pattern that is observed. [2]

iv. In another experiment using the same apparatus, a student notices that the angular separation between the zero order maxima and the two 1st order maxima are not equal.

Suggest a reason for this difference. [1]

[Total: 20]

12. [PJC/Prelims 2014/P2/3]

Fig. 3.1 shows a transmitter T on a building, which emits radio waves of wavelength 2.0×10^3 m. The radio waves from transmitter T travels to a receiver R by two paths as shown in Fig. 12.1 (not to scale).

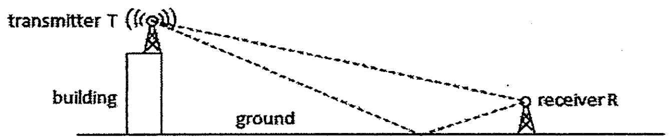


Fig. 12.1

For the first path, the wave travels 2.4×10^4 m directly from T to R. For the second path, the wave travels to the ground and is reflected to R. Maximum intensity is received at R. It is assumed that there is no change in phase of the radio wave upon reflection at the ground.

a. Explain why maximum intensity is received at R. [2]

b. Calculate the minimum distance travelled by the radio wave in the second path (reflected path), which will produce maximum intensity at R. [3]

c. In order to receive radio waves, an antenna must be used at R.

i. Calculate the frequency of the radio waves received at R. [2]

ii. The antenna is configured to resonate at a particular frequency.

Suggest a reason why the phenomenon of resonance is used in receiving radio wave. [1]

[Total: 8]

13. [SAJC/Prelims 2014/P3/6 Part]

c. Coherent light of wavelength 590 nm is incident normally on a double slit as shown in Fig 13.1.

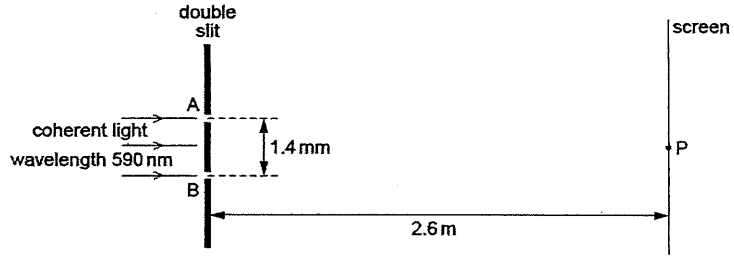


Fig. 13.1

The separation of slits A and B is 1.4 mm.

Interference fringes are observed on screen placed parallel to the plane of the double slit. The distance between the screen and the double slit is 2.6 m.

A point P on the screen, the path difference is zero for light arriving at P from the slits A and B

i. Determine the separation of bright fringes on the screen near to point P. [2]

ii. Variation with time of the displacement x of the light wave arriving at point P on the screen from slit A and from slit B is shown in Fig. 13.2 and 13.3 respectively.

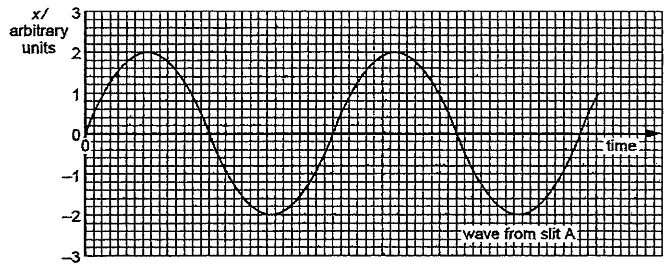


Fig. 13.2

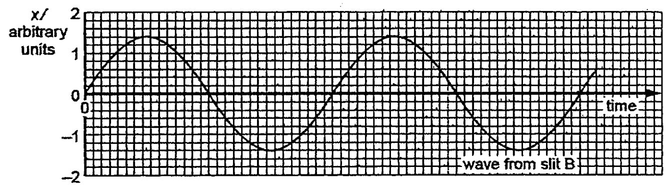


Fig. 13.3

1. State the phase difference between waves forming the dark fringe on the screen that is next to point P. [2]

2. Determine the ratio $\dfrac{\text{intensity of light at a bright fringe}}{\text{intensity of light at a dark fringe}}$. [3]

d. The double-slit in Fig. 13.1 is replaced by a diffraction grating with 650 lines per millimetre, as shown in Fig. 13.4.

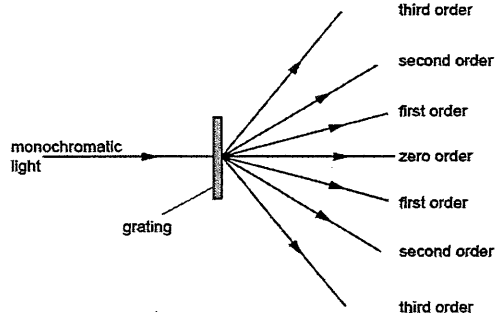


Fig. 13.4

i. For incident light of wavelength 590 nm, determine the number of orders of diffracted light that can be observed on each side of the zero order. [2]

ii. Explain why the interference pattern produced by light passing through a diffraction grating is brighter than that produced from the same source with a double slit. [1]

iii. In what way(s) would the fringe pattern change if the monochromatic light was replaced by a source producing white light? Briefly explain. [2]

[Total: 12]

14. [SRJC/Prelims 2014/P2/5]

a. Describe the formation of a stationary wave. [1]

b. A stretched string is used to form a stationary wave. Part of this wave, at a particular instant, is shown in Fig. 14.1.

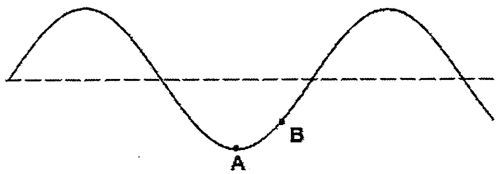


Fig. 14.1

The points on the string are at their maximum displacement.

i. State the phase difference between the particles labelled A and B. [1]

ii. State the number of antinodes shown in Fig. 14.1 for this wave. [1]

iii. The period of vibration of this wave is T. On Fig. 5.1, sketch the stationary wave $0.25T$ after the instant shown in Fig. 14.1. [1]

[Total: 4]

15. [SRJC/Prelims 2014/P3/7]

a. State the Principle of Superposition. [2]

b. Monochromatic light illuminates a narrow slit which is 4.0 m away from a screen. Two very narrow parallel slits 0.50 mm apart are placed midway between the single slit and the screen so that interference fringes are obtained.

i. Explain how diffraction and interference play a part in formation of the interference fringes. [2]

ii. The spacing of five bright fringes is 10 mm, calculate the wavelength of the light. [2]

iii. State the changes in the fringes, if any, when the screen is moved closer to the double slit. [2]

iv. The intensity of the bright fringe is I. A very thin paper is placed in front of one of the double slits such that the intensity of the light passing through the slit is now halved. Determine the intensity, in terms of I, of the new minima. [3]

v. The monochromatic light source is now replaced by a light source producing red and blue light. State and explain what will be observed at the central fringe. [2]

c. Explain why

i. when the light beams from the two headlamps of a car overlap, it is not possible to have an observable interference pattern. [1]

ii. when a source producing sound waves of wavelength 5.0 m is connected to two loudspeakers which are placed 2.0 m apart, it is not possible to locate a position where there will be a minima. [1]

d. A monochromatic light of wavelength 700 nm is incident on a diffraction grating that has 2750 lines per centimeter. The second order spectrum appears at a distance of 3.0 cm away from the zeroth order spectrum as shown in Fig. 15.1.

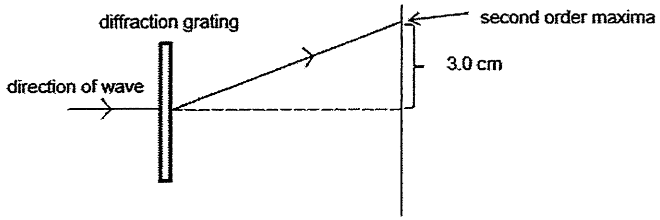


Fig. 15.1

i. Determine how far the screen is away from the diffraction grating. [3]

ii. When a patient receives a chest x-ray at the hospital, the x-rays pass through a series of parallel ribs in the patient's chest.

Explain whether the ribs act as a diffraction grating for x-rays. [2]

[Total: 20]

16. **[TJC/Prelims 2014/P3/6]**

a. Explain what is meant by interference **[2]**

b. State two conditions which must be satisfied in order to have observable interference fringes. **[2]**

c. If two wave sources are in phase, state the condition imposed on the path difference for the waves to interfere constructively at a point. **[1]**

d. Two microphones, M_1 and M_2, are positioned at a distance apart as shown in Fig. 16.1. The signals from the microphones are added and the resultant signal is displayed on a CRO. A speaker **S** which is producing sound waves of constant frequency and in all directions is placed along a line joining the microphones.

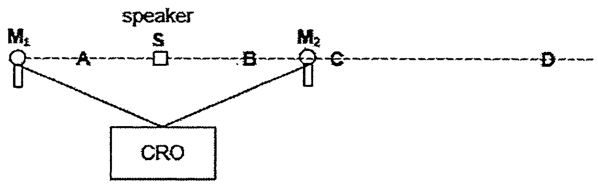


Fig. 16.1

i. When the speaker is moved from point A to B, the amplitude of the resultant waveform shown on the CRO goes through a series of maxima and minima. Explain the formation of the maxima and minima. **[2]**

ii. Later, the speaker is placed at C and moves towards D. Explain why the amplitude of the resultant waveform does not fluctuate. You may assume that the intensity of the wave did not drop significantly with distance from the speakers. **[2]**

e. In another experiment, a microphone **M** is placed in between 2 identical speakers S_1 and S_2 as shown in Fig. 16.2. A signal generator now drives the speakers to produce sound waves of constant frequency 3.0 kHz.

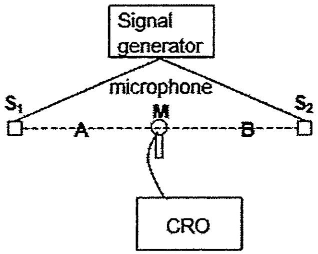


Fig. 16.2

i. When the microphone M is moved from A to B at a speed of 10.0 cm s^{-1}, the amplitude of the waveform on the CRO goes through a series of maxima and minima at a rate of 2 maxima per second.

Calculate the speed of sound. **[2]**

ii. Explain why in practice, the amplitude shown on the CRO at most minima points is not zero. **[2]**

f. Fig. 16.3 shows an experimental arrangement for the study of transmission of light through a diffraction grating which has 600 lines per millimetre and is situated 2.0 m from the screen.

A 400 nm monochromatic light from source S is shone normally on the grating. The first order bright fringe is observed at a distance x from the central bright fringe.

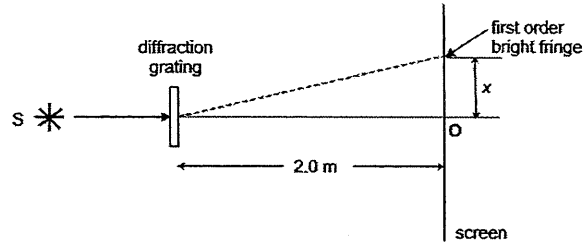


Fig. 16.3

i. Determine the distance x. **[3]**

ii. Calculate the maximum number of fringes that can be seen on the screen. **[2]**

iii. When a different diffraction grating is used, the first order bright fringe is found to be at a distance $2x$ from the central bright fringe. Determine the number of lines per mm for this new grating. **[2]**

[Total: 20]

17. **[TPJC/Prelims 2014/P2/3 Part]**

b. Fig. 17.1 shows an arrangement which demonstrates interference of sound waves.

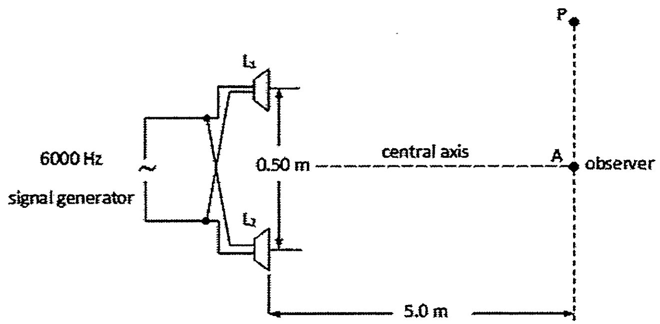


Fig. 17.1

The two loudspeakers L_1 and L_2 are driven by the same signal generator set to a frequency of 6000 Hz. The speakers emit the same power at this frequency. An observer identifies A as being the position where the loudest sound is heard.

i. Given that A is the position where loudest sound is heard, state and explain what conclusions can be drawn concerning the phase difference of the sound waves emitted by the two loudspeakers. **[3]**

ii. Estimate how far the observer would need to move towards P in order to be at a position where a minimum sound intensity would be detected. Speed of sound in air = 340 m s^{-1}. **[3]**

[Total: 6]

18. [TPJC/Prelims 2014/P2/8 Part]

a. Explain what is meant by diffraction of waves. **[1]**

b. A double slit with slit separation 0.800 mm is situated a distance 2.50 m from a thin jet of high speed smoke as shown in Fig. 18.1.

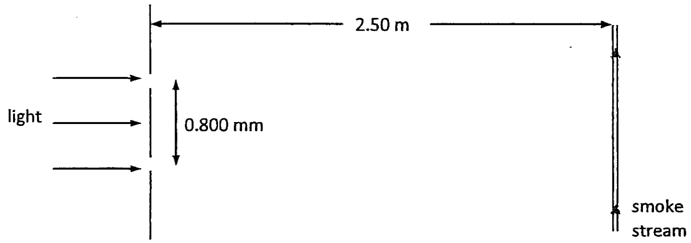


Fig. 18.1

i. The double slit is illuminated with coherent light of wavelength 589 nm. Fringes are observed in the moving smoke.

Calculate the separation of these fringes. **[2]**

ii. Suggest and explain a change to the appearance of the fringes when each of the following changes is made separately.

1. The direction of the smoke stream is rotated through 45° as shown in Fig. 18.2. **[2]**

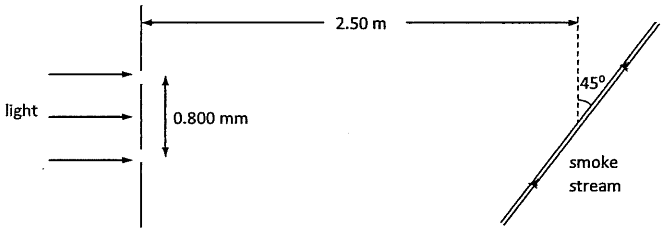


Fig. 18.2

2. The double slit in Fig. 18.1 is replaced with a crossed double slit (i.e. a combination of a vertical and a horizontal double slit). **[2]**

c. A slit allows a narrow beam of white light to emerge from it and fall on a diffraction grating at normal incidence. Wavelengths of the light range from 430 nm to 680 nm.

It is found that the first order spectrum is displaced from the straight-through position at an angle of 28° and this spectrum has an angular width of α as shown in Fig. 18.3.

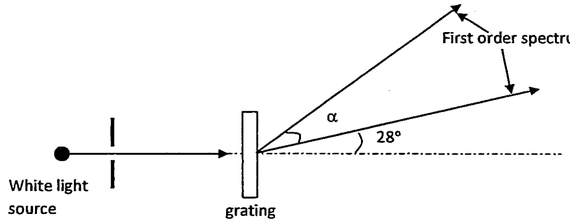


Fig. 18.3

i. Calculate the number of lines per centimetre of the diffraction grating. **[3]**

ii. Hence determine the angular width, α, in degree, of the first order spectrum. **[2]**

[Total: 12]

19. [VJC/Prelims 2014/P3/5 part]

b. Fig. 19.1 shows a full-scale photograph of an interference pattern produced when monochromatic light falls on a pair of slits 0.30 mm apart. The pattern was produced on a screen 1.5 m from the slits.

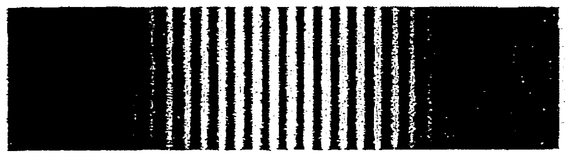


Fig. 19.1

i. State the principle of superposition and use this principle to explain why such an interference pattern is produced when light falls on a pair of closely spaced slit. **[3]**

ii. Use the photograph to obtain a value for the fringe spacing. Show your working clearly. **[2]**

iii. Calculate the wavelength of the light used. **[2]**

iv. Explain why the fringes near the centre of the photograph are brighter than those near the edges of the photograph. **[2]**

v. Sketch the pattern which would be obtained on the screen if one of the slits were covered up. **[1]**

[Total: 10]

20. [YJC/Prelims 2014/P2/3]

a. Describe what is meant by diffraction. **[1]**

b. In an experiment, a monochromatic light is incident normally on a double slit, and the interference pattern is observed on a screen situated a distance *D* from the slits. The fringe spacing *x* is measured. The experiment is repeated with different values of *D* and corresponding values of *x* are obtained.

Fig. 20.1 shows a graph of the variation of *x* with *D*.

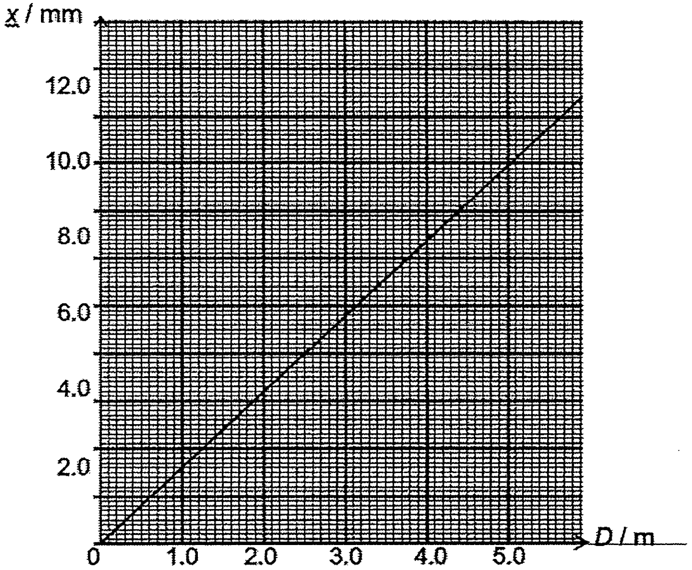


Fig. 20.1

i. State what is meant by the term *monochromatic*. **[1]**

ii. Determine the gradient of the graph. **[1]**

iii. Using your answer in **ii.**, calculate the spacing between the slits, given the wavelength of light is 620 nm. **[2]**

iv. If the experiment is repeated with light of a longer wavelength, state, if any, the changes to the graph. **[1]**

[Total: 6]

21. [YJC/Prelims 2014/P3/4]

A narrow slit allows a narrow beam of white light to emerge from it and incident on a diffraction grating as shown in Fig. 21.1.

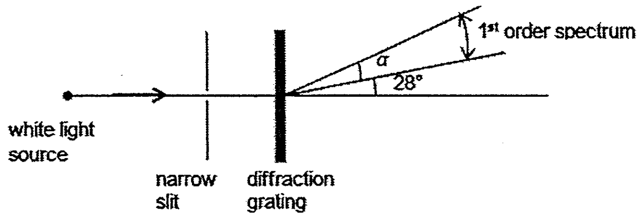


Fig. 21.1

The beam contains wavelengths which vary from 430 nm to 680 nm. The first order spectrum is displaced from the central fringe by an angle of 28° and it spans at an angle of α.

a. Determine

i. the number of slits per metre on the diffraction grating **[2]**

ii. the angle α. **[1]**

b. State and explain the appearance of the central fringe. **[2]**

c. At higher orders of the spectrum, there will be overlapping of certain wavelengths.
Explain the term *overlapping*'. **[1]**

d. If the white light is replaced by a monochromatic source of wavelength 600 nm, determine the maximum number of bright lines that will be seen on the screen. **[2]**

[Total: 8]

Raffles Institution Physics Department Revision Package 2	Superposition	11

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	A	A	B	B	C	C	D	A	B	C	C	A	B	D

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	B	A	C	A	B	D	B	D	C	D	D	C	B	D

31	32	33	34	35	36	37	38	39	40	41
A	D	C	D	B	D	B	C	B	D	B

1 A The standard double slit interference pattern is shown in A, with equal intensities for the peaks that are evenly spaced. Notice that the fringes are not enveloped by single-slit diffraction pattern, since the fringes are formed by two sources (no single-slit diffraction is involved).

2 A The upper (horizontal) path is of length 40 cm, whereas the lower path is of length $\sqrt{40^2 + 9^2} = 41$ cm. Path difference of 1 cm, which is half a wavelength, implying destructive interference at X.

3 A Fringe separation, x = λD/d (assuming small angle approximation applies), where λ is the wavelength of the light, D is the distance between slits and screen and d is the slit separation.

Options C and D reduces the fringe separation while option A increases fringe separation (as the wavelength of orange light is larger).

Option B has no effect on fringe separation.

4 B The distance between the first loud sound and the 6th loud sound is 5 segments, the length of each segment being equal to half a wavelength.

Distance between adjacent nodes = (37 – 4.0) / 5 = 6.6 cm

Wavelength of sound = 2 × 6.6 = 13.2 cm

Speed of sound = 2500 × 0.132 = 330 m s^{-1}

5 B Using $v = f\lambda$, $\lambda = 320 / 400 = 0.80$ m

No. of wavelengths in tube = 4.0 / 0.80 = 5

No. of antinodes in 5 wavelengths = 11

If frequency is doubled, $\lambda = 320 / 800 = 0.40$ m

No. of wavelengths in tube = 4.0 / 0.40 = 10

No. of antinodes in 10 wavelengths = 21

Thus, there are 10 more antinodes in the stationary wave.

6 C Option C is true as the maximum intensity point will not shift although the maximum intensity will decrease.

Option A is wrong because a single-slit diffraction pattern will be seen.

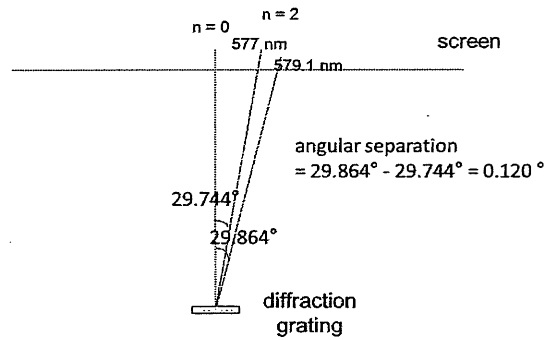
Option B is wrong because the fringe separation is not dependent on the distance between light source and slit.

Option D is wrong because replacing the laser light with one of higher wavelength will cause the fringe separation to increase but will not shift the central bright fringe (0th order maximum).

7 C $d = \dfrac{10^{-2}}{4300} = 2.326 \times 10^{-6}$ m

Using $a \sin\theta = n\lambda$ for both wavelengths 577.0 nm and 579.1 nm for the different orders will give the following:

n	θ / ° for 577.0 nm	θ / ° for 579.1 nm	$\Delta\theta$ / °
2	29.744	29.864	0.120
3	48.090	48.323	0.233
4 (max order)	82.867	84.793	1.926
5	NA	NA	NA



8 D Points in the same segment (i.e. between consecutive nodes) vibrate with different amplitudes but are in phase; points in neighbouring segments vibrate in antiphase.

9 A $a\sin\theta = n\lambda \Rightarrow \dfrac{1}{N}\sin\theta = n\lambda \Rightarrow \sin\theta = Nn\lambda \Rightarrow$ gradient = Nn

10 B The double-slit interference is between the loudspeaker and its 'mirror image', where the wall is the 'mirror'. Consider the equation $a\sin\theta = n\lambda$, where a is the slit-separation (equal to twice the distance between the loudspeaker and the wall). In order to increase the distance between loud and soft sounds, one needs to increase θ, which is the angular positions of the constructive interference (loud sounds).

Option A: To increase frequency is to decrease wavelength. According to the equation, θ decreases.

Option B: Moving the loudspeaker closer to the wall amounts to decrease a, which increases θ (since the product $a\sin\theta$ remains unchanged, since λ didn't change).

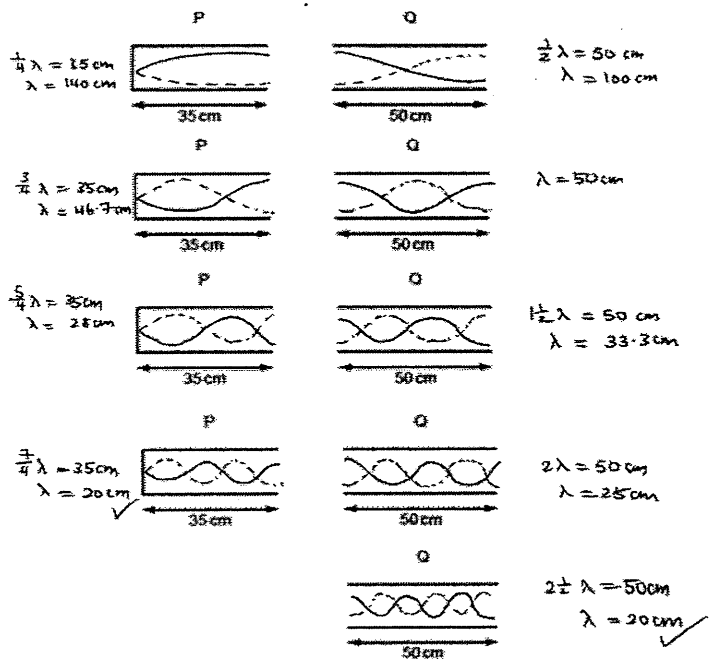
Option C: Moving the loudspeaker towards L does not affect the angular position of the loud sounds. The linear separation between loud sound, however, decreases ($\Delta x = \frac{\lambda D}{a}$, where D is the distance between S and L).

Option D: No effect.

11 C The light emitted by each lamp undergoes independent and abrupt changes in phases. Hence the phase difference between the two light sources are not constant in time.

12 C Using $v = f\lambda$, wave velocity 153.5 m s^{-1} and frequency 256 Hz correspond to wavelength 0.60 m. Hence both the 1st harmonic ($L = \frac{\lambda}{2}$) on the 30 cm-string and the 2nd harmonic ($L = \lambda$) on the 60 cm-string can cause the tuning fork to vibrate.

13 A



Standing waves formed in P has a node on the left end and an antinode on the right end. Hence, the length of the pipe L must satisfy $L = \left(\frac{n}{2} + \frac{1}{4}\right)\lambda$, where n is an integer. Using $L = 35$ cm and $\lambda = 20$ cm, n is found to be 3. Hence, standing wave can be formed in pipe P.

Similarly, standing wave formed in Q has antinodes on both ends, and the pipe length L must satisfy $L = \frac{n}{2}\lambda$. Using $L = 50$ cm and $\lambda = 20$ cm, n is found to be 5. Hence, standing wave can be formed in pipe Q as well.

14 B From the lower diagram, the linear separation between the bottom (at 15 mm) and the top (at 30 mm) bright fringes is 15 mm. This distance is equivalent to 5 fringe separations (as can be counted on the diagram).

Assuming small-angle approximation applies (since the fringes appear to be evenly spaced):

$$x = \frac{\lambda D}{a}$$

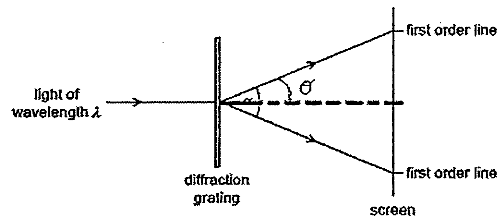
$$\frac{15\times10^{-3}}{5} = \frac{\lambda(5.0)}{0.9\times10^{-3}}$$

$$\lambda = 5.4\times10^{-7}\text{m}$$

15 D $a\sin\theta = n\lambda$

$$\theta = \sin^{-1}\left(\frac{\lambda}{d}\right) \quad \text{for } n = 1$$

$$\alpha = 2\theta = 2\sin^{-1}\left(\frac{\lambda}{d}\right)$$



16 C The amount of diffraction depends on the size of the opening relative to the wavelength.

17 B No energy is transferred in a stationary wave.

Options A and C are obviously wrong.

Option D is wrong because particles at nodes in a standing wave do not vibrate.

18 A From a node, through 15 antinodes, to another node is equivalent to 15 × ½ wavelengths.

$$\lambda = \frac{1.200}{15/2} = 0.160 \text{ m}$$

On the CRO, 5 consecutive crests represents 4 periods.

$$T = \frac{8.5}{4}\times\left(0.50\times10^{-3}\right) = 1.0625\times10^{-3} \text{ s}$$

$$v = f\lambda = \frac{1}{T}\lambda = 150 \text{ m s}^{-1}$$

19 C $a\sin\theta = n\lambda$

$$\frac{a}{\lambda} = \frac{n}{\sin\theta} = \frac{2}{\sin(32°)} = 3.7$$

Hence max order is 3, and 3 + 1 + 3 = 7 bright spots can be seen (including the zeroth order).

20 A The incident sound wave and the reflected sound wave from the wall form a standing wave, with a displacement node at the wall. The distance between neighbouring node and antinode is $\frac{1}{4}\lambda$, whereas the distance between neighboring antinodes is $\frac{\lambda}{2}$.

Using $v = f\lambda$, it is easy to deduce that $\lambda = 3.40$ m, $\frac{\lambda}{4} = 0.85$ m and $\frac{\lambda}{2} = 1.70$ m.

21 B

$$\lambda = \frac{v}{f} = \frac{3.0\times10^{8}}{1.5\times10^{6}} = 200 \text{ m}$$

Hence the distance between successive nodes is 100 m (half a wavelength). That is to say, the ship receives a strong signal (antinode) every 100 m (since there is one antinode between successive nodes).

Thus, the ship receives a strong signal every (100/2.0 =) 50 seconds, since the ship travels at 2.0 m s^{-1}. Hence, the frequency is 1/50 = 0.02 Hz.

22 D From $a\sin\theta = n\lambda$, larger wavelength will be diffracted at larger angles.

Hence first order violet fringes are closest to the central maximum (white fringe at the centre), which rules out Option A.

The second order violet fringe is determined by $d\sin\theta_v = 2\times400$, whereas the first order red fringe is determined by $d\sin\theta_r = 700$, thus $\theta_v > \theta_r$, and there is a distance between them. This rules out Options B and C.

23 B The distance between X and Y (two successive loud sounds) is half a wavelength, hence $\lambda = 66$ cm. Using $v = 330$ m s^{-1}, period $T = \frac{\lambda}{v} = 0.002$ s $= 2$ ms.

24 D No resonance is found at any lower frequency than f with tube X, which is a closed pipe, implying that the correspond wavelength satisfies $L = \lambda/4$, or $\lambda = 4L$.

To form a standing wave (resonance) in tube Y, which is open, the wavelength must satisfy $L = n\lambda'/2$, where L is the same length as X, or $\lambda' = 2L/n$, n being an integer.

Option A: $f/2$ implies wavelength $\lambda' = 2\lambda$, or $\frac{2L}{n} = 4L$ (using $\lambda' = 2L/n$ and $\lambda = 4L$). No integer n satisfy this equation.

Option B: $2f/3$ implies wavelength $\lambda' = 3\lambda/2$, or $\frac{2L}{n} = 6L$. No integer n satisfy this equation.

Option C: $3f/2$ implies wavelength $\lambda' = 2\lambda/3$, or $\frac{2L}{n} = \frac{8L}{3}$. No integer n satisfy this equation.

Option D: $2f$ implies wavelength $\lambda' = \lambda/2$, or $\frac{2L}{n} = 2L$, which implies that $n = 1$.

25 C Coherence is essential for interference to occur.

26 D $I = kA^2$, k being a constant.

For maximum intensity, amplitude $= 2A + A = 3A$, $I_o = k(3A)^2$

For minimum intensity, amplitude $= 2A - A = A$, $I = k(A)^2$

$$\frac{I}{I_o} = \left(\frac{A}{3A}\right)^2 \Rightarrow I = \frac{I_o}{9}$$

27 D

$$a\sin\theta = n\lambda$$

$$a\sin\left(\frac{166° - 140°}{2}\right) = 2(500\times10^{-9})$$

$$a = 4.45\times10^{-6} \text{ m}$$

28 C

The stationary wave formed in the tube has a displacement node at the closed end. X is a distance $\frac{3}{4}\lambda$ from the node, hence is a displacement antinode, and hence a pressure node.

29 B For destructive interference, path difference = $(2n + 1)\lambda/2 = (P_3 + P_4) - (P_2 + P_1)$

30 D

$$a\sin\theta = n\lambda$$

$$\sin 7.4° = \frac{1(\lambda)}{d}$$

For 3rd order diffraction, $\theta_3 = sin^{-1}\left(\frac{3(\lambda)}{d}\right) = 22.7°$

θ_3 = angle between 3rd order maximum and the incident wave.

Between the two 3rd order diffraction maxima, $\theta = 2\times\theta_3 = 45.4°$

31 A Points within the same segment are in phase; points in neighbouring segments are in antiphase.

32 D A and B are in neighbouring segments of a standing wave, hence are in antiphase. B is an antinode, hence vibrates with maximum amplitude (max KE is proportional to amplitude squared).

33 C For constructive interference, path difference should be $(n + ½)\lambda$ (taking into account the 180 degree phase change at reflection).

For longest wavelength, n should be 0.

Path taken by the wave that rebounded from the ground is $\sqrt{(400^2 + 200^2)} + \sqrt{(300^2 + 600^2)}$ = 1118 m,

Path taken by wave that reach the receiver directly is $\sqrt{(200^2 + 500^2)}$ = 539 m

Hence path difference is 1,118 – 539 = 579 m.

$½\lambda = 579 \Rightarrow \lambda = 1160$ m

34 D Waves travel in opposite directions for PQ; waves intersecting with different path differences along XY; waves have constant path difference of zero along RS.

35 B $\Delta x = \lambda D/a = (6.0 \times 10^{-7} \times 1.50) / (3.0 \times 10^{-3}) = 0.30$ mm

36 D $I = kA^2$, k being a constant.

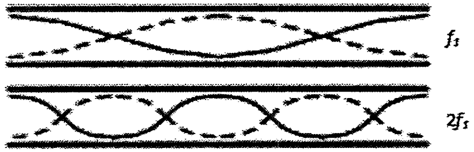
M is at the centre of the screen, where the two slits interfere constructively when both are in use, amplitude $= A + A = 2A$, $I_0 = k(2A)^2$

When only 1 slit is in use, amplitude $= A$, $I = k(A)^2$

$$\frac{I_0}{I} = \left(\frac{2A}{A}\right)^2 \Rightarrow I_0 = 4I$$

37 B Points in the same segment (i.e. between consecutive nodes) vibrate with different amplitudes but are in phase; points in neighbouring segments vibrate in antiphase.

38 C Twice the frequency implies half the wavelength. By drawing the stationary waves that has twice the frequency of f_1, it can be counted that there are 5 antinodes.



39 B Waves from P has intensity I and amplitude A.

Waves from Q has intensity 4 I and thus has amplitude $2A$.

Path difference = PX – QX = 2.5 – 2.0 = 0.5 m

Since wavelength is 0.04 m, then the path difference = 0.5 / 0.04 = 12.5λ.

Thus at point X the waves meet out of phase.

Resultant amplitude = $2A - A = A$.

Hence resultant intensity = I.

40 D Consider the equation $a \sin\theta = n\lambda$, where a is the slit-separation. In order to increase the fringe separations, one needs to increase θ, which is the angular positions of the constructive interference. If small angle approximation is applicable, also consider $\Delta x = \frac{D\lambda}{a}$, where D is the distance between the slits and the screen.

Option A: Decreasing D does not affect the angular positions of the fringes. The linear separation, however, decreases (according to $\Delta x = \frac{D\lambda}{a}$).

Option B: Increasing d decreases $\Delta\theta$ (since the product $a \sin\theta$ remains unchanged, since λ didn't change).

Option C: No effect.

Option D: Red light has larger wavelength. Increasing wavelength increases θ according to $a \sin\theta = n\lambda$.

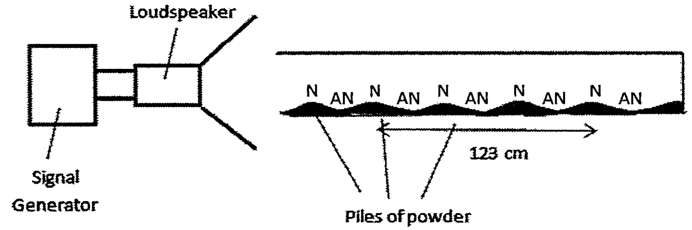
41 B The path difference (4.0 m) is equal to 8 wavelengths. Since S_1 and S_2 are in antiphase, the two waves interfere destructively at P. Hence, the resulting amplitude $= 3a - a = 2a$.

Structured Questions

1. [ACJC/Prelims 2014/P2/4]

a. See definitions in lecture notes.

i.



ii. When the sound wave that are produced by the source is reflected at the glass boundary with a phase difference of π rad.

The incident and reflected waves interfere. As they have the speed and frequency, this results in the formation of nodes and antinodes of a standing wave if the length of the tube is given by odd multiples of $\lambda/4$, where λ is the wavelength of the sound wave.

The piles of power gather at the displacement nodes as at the nodes, no oscillations of air particles occur.

iii. wavelength of stationary wave = 123 × 2/3 = 82 cm

speed of wave = 400 × 0.82 = 328 m s^{-1}

2. [ACJC/Prelims 2014/P3/5]

a. i.

$$ax = \lambda D$$

$$(0.400 \times 10^{-3})(3 \times 10^{-3}) = \lambda \times (3.2)$$

$$\lambda = 3.75 \times 10^{-7} \text{ m}$$

ii. $\frac{\text{path difference}}{0.400 \times 10^{-3}} = \frac{5.0 \times 10^{-3}}{3.2} \Rightarrow$ path difference $= 6.25 \times 10^{-7}$ m

phase difference $= \frac{\text{path difference}}{\lambda} \times 2\pi = \frac{6.25 \times 10^{-7}}{3.75 \times 10^{-7}} \times 2\pi = 3.3\pi$

b. $\frac{\text{additional path difference}}{0.400 \times 10^{-3}} = \frac{1.0 \times 10^{-3}}{3.2} \Rightarrow$ additional path difference $= 1.25 \times 10^{-7}$ m

$(n-1)(1.0 \times 10^{-6}) = 1.25 \times 10^{-7} \Rightarrow n = 1.13$

3. [AJC/Prelims 2014/P2/3]*

a. See definitions in lecture notes.

b. i. Angle of diffraction of the first order = $\tan^{-1}(10/30) = 18.43°$

$a \sin\theta = n\lambda \Rightarrow a = (1)(580 \times 10^{-9}) / \sin(18.43°) = 1.834 \times 10^{-6}$ m

number of lines per mm = $(1 \times 10^{-3}) / (1.834 \times 10^{-6}) = 545$ lines

ii. $E = hf$

$[-2.71 - (-5.77)] \times 1.6 \times 10^{-19} = 6.63 \times 10^{-34} \times 3.0 \times 10^{8} / \lambda$

$\lambda = 4.06 \times 10^{-7}$ m

iii. To determine the highest order, $\sin\theta \le 1$.

$a \sin\theta = n\lambda \Rightarrow n\lambda / a \le 1 \Rightarrow n \le 1.834 \times 10^{-6} / 4.0625 \times 10^{-7} \Rightarrow n \le 4.51$

highest order = 4

4. [CJC/Prelims 2014/P2/2 Part]

c. i. Using $x = \dfrac{\lambda D}{a}$ where $\lambda = \dfrac{v}{f}$

$y = \dfrac{(\frac{330}{1000})(15)}{(1.2 \times 2)}$, x is twice that of y ($x = 2y$) since B is at minimum intensity

$y = 2.1$ m

ii. Since B is a point of minimum intensity, the path difference of S_1 and S_2 at point B,

$S_1B - S_2B = \dfrac{\lambda}{2} = \dfrac{0.33}{2} = 0.165\ m$

For other wavelengths λ, point B will be at zero intensity if the path difference is odd multiple of $\dfrac{\lambda}{2}$.

$(2n+1)\dfrac{\lambda}{2} = 0.165$ m where $n = 1 \Rightarrow \lambda = 0.11$ m

Using $f = \dfrac{v}{\lambda}$, $f = \dfrac{330}{\lambda} = 3000$ Hz

iii. Since amplitude $\propto$ voltage and amplitude$^2 \propto$ intensity,

initially at A, constructive interference occurs as the two waves meet in phase to give I_{max}.

Thus amplitude proportional to 6 V + 6 V = 12 V, and $I_{max} \propto (12\text{ V})^2 = 144\text{ V}^2$

At B, destructive interference occurs as two waves meet in antiphase,

Thus amplitude proportional to 6 V – 6 V = 0 V giving $I_{min} = 0$.

When the speakers are adjusted to 9 V and 3 V,

the resultant amplitude at B is proportional to 9 V – 3 V = 6 V.

Hence $I_{min} \propto (6\text{ V})^2 = 36\text{ V}^2$

$I_{min} = \frac{1}{4} I_{max}$

5. [DHS/Prelims 2014/P2/2]

a. waves must meet

waves must be of the same (transverse/longitudinal) type

waves must be polarized in the same plane or both unpolarized

b. i. to produce coherent sources at the two slits (S_1 and S_2)

ii. intensity = k(amplitude)^2 → amplitude of each wave = $\sqrt{\dfrac{I}{k}}$

resultant intensity = k(resultant amplitude)$^2 = k\left[\sqrt{\dfrac{I}{k}} + \sqrt{\dfrac{I}{k}}\right]^2 = 4I$

iii. π rad OR 180°

iv. light from s_1 and s_2 interferes destructively, leaves light from s_3 at P, P not dark

6. [HCI/Prelims 2014/P2/3]

a. i. Microwaves.

The degree of diffraction of a wave depends on its wavelength relative to the opening/obstacle size.

ii. $\lambda = c / f = (3.00 \times 10^{8}) / (90.5 \times 10^{6}) = 3.315$ m

400 nodes correspond to 399 $\lambda/2$ segments.

distance is 399 × 3.315/2 = 661 m

b. i. They must be waves of the same type.

They must overlap.

ii. $\Delta y = \lambda D / a$

$\Delta y = (690 \times 10^{-9}) \times (1.20) / (0.333 \times 10^{-3}) = 0.002486$ m = 2.49 mm

iii. Since the screen is 2.40 m wide and a distance 1.20 m away, the maximum diffraction angle is 45°.

$a \sin\theta = n\lambda$

$n = d \sin\theta / \lambda$

$n = (1 \times 10^{3} / 300) \sin 45° / (690 \times 10^{-9}) = 3.4$

The highest visible order is 3

7. [IJC/Prelims 2014/P3/4]

a. See definitions in lecture notes.

b. To produce 2 water waves with circular wavefronts, 2 (ball-typed) dippers are used.

To ensure that they are coherent, the dippers are connected to the same vibrator source / motor.

OR 1 dipper as source and 2 slits

For observations of interference patterns, a lamp is set-up above the tank and a screen below the tank.

c. i. $v = f\lambda \Rightarrow 320 = 400 \times \lambda \Rightarrow \lambda = 0.800$ m

ii. Distance SPX is 7.0 m and distance SX is 5.0 m, hence the path difference is 2.0 m which is equivalent to 2.5 λ. It is a maximum.

After considering the extra π radians of phase change upon reflection. The 2 waves meet in phase.

8. [JJC/Prelims 2014/P3/8 Part]

b. Resultant amplitude, $A_R = 3A - A = 2A$

$I = kA^2 \Rightarrow$ resultant intensity $I_R = kA_R^2 = k(2A) = k(4A^2) = 4I$

c. i. As the microphone is moved between the loudspeaker and the reflector, mark a position where the CRO attached shows the maximum (or minimum) amplitude.

Then move the microphone along to the next position where the amplitude shown is maximum (or minimum) again.

The distance between these two positions is half a wavelength.

ii. When microphone detects a maximum, its diaphragm vibrates with maximum amplitude. This means that the air pressure there has a maximum change. It implies that that position is alternatively a compression and a rarefaction, which is a displacement node.

d. i. See definitions in lecture notes.

ii. When a wave front meets the pair of slits, the wave front emerges from the slits as divergent wave fronts which are in phase with each other, i.e. the two slits act as two coherent sources of divergent waves which are in phase. Hence the coherence condition for interference is obtained.

e. $x = \dfrac{\lambda D}{a} = \dfrac{(5.24\times10^{-7})(0.945)}{0.220\times10^{-3}} = 2.25\times10^{-3}$ m

f. i. $\sin\theta = \dfrac{n\lambda}{a} \Rightarrow \lambda = \dfrac{a\sin\theta}{n} = \dfrac{(2.20\times10^{-6})\sin 11.8°}{1} = 4.50\times10^{-7}$ m

ii. $\sin\theta = \dfrac{n\lambda}{a} \Rightarrow n_{violet} = \dfrac{a\sin\theta}{\lambda_{violet}} = \dfrac{(2.20\times10^{-6})\sin 54.9°}{4.50\times10^{-7}} = 4$

$n_{red} = \dfrac{a\sin\theta}{\lambda_{red}} = \dfrac{(2.20\times10^{-6})\sin 54.9°}{5.99\times10^{-7}} = 3$

i.e. at 54.9°, both maxima of violet and red light overlap.

9. **[MJC/Prelims 2014/P3/6 Part]**

a. i. 1. See definitions in lecture notes.

2. See definitions in lecture notes.

3. See definitions in lecture notes.

ii. Sources of waves must be coherent

Waves must have equal or similar amplitudes

waves must be polarized in the same plane or both unpolarized for transverse waves

waves must meet / waves must be of the same (transverse/longitudinal) type

b. i. Since the two radio waves are in phase, along centre-line, path difference is always zero.

Hence constructive interference occurs.

ii. Radio waves have long wavelengths. Hence, the anti-nodal lines will be far apart enough for the ship to differentiate.

iii. Since the intensity of each individual wave is inversely proportional to the square of the distance, the intensity of each individual wave will increase as the ship goes nearer, hence the resultant intensity will increase.

OR

Since the amplitude of each individual wave is inversely proportional to distance, the amplitude of each individual wave will increase as the ship goes nearer, hence the resultant amplitude of the superposed wave will increase. As intensity is proportional to the square of the amplitude, intensity increases.

Hence, the intensity of the resultant increases as the ship approached the gate.

c. $\lambda_1 = \dfrac{c}{f_1} = \dfrac{3.0\times10^8}{23.5\times10^6} = 12.77$ m

Path difference $= \sqrt{(895)^2 + \left(250+\dfrac{95}{2}\right)^2} - \sqrt{(895)^2 + \left(250-\dfrac{95}{2}\right)^2} = 25.527 \text{ m} = 2\lambda_1$

Since the path difference is approximately $2\lambda_1$, the ship is on an anti-nodal line.

Note: using $x = \dfrac{\lambda D}{a}$ will earn no credit.

d. i. If ship is on the central anti-nodal line, it should detect maximum signals from both frequencies / the maximum signal will be stronger.

OR

If ship is on wrong anti-nodal line, only one of the frequencies will show a strong signal.

ii. Higher orders of maxima from both frequencies may still coincide/overlap.

Hence the ship could still detect maximum signals from both frequencies even though it is not on the central anti-nodal line.

10. **[NJC/Prelims 2014/P3/7 Part]**

d i. It is an effect that occurs when the 2 coherent wave trains overlap to produce a new wave pattern. Destructive interference will be observed if the 2 wave trains are π radian out of phase with one another. Constructive interference will occur when the 2 wave trains are in phase with one another.

ii. 1. Fringe separation, $x = \lambda D/a = 6.0\times10^{-7}\times0.80 / 0.5\times10^{-3} = 9.6\times10^{-4}$ m

2. The speed of light decreases inside the glass. This created an extra path difference (optical path difference). To make up the extra path difference S_1P must cover additional distance. Hence S_1P must be longer than before. Hence P is shifted downwards.

OR

Since the speed decreases, the number of wavelengths increases inside the glass. Hence S_1P must cover extra distance (number of wavelengths) to make up for the difference. Hence P will be shifted downwards.

3. Wavelength of light in glass = $6.0\times10^{-7} / 1.5 = 4.0\times10^{-7}$ m

4. Increase in number of waves in G = $(3.6\times10^{-6} / 4.0\times10^{-7}) - (3.6\times10^{-6} / 6\times10^{-7}) = 9 - 3 = 3$

5. Path difference due to G = 3λ

Length of OP = $3(\lambda D/a) = 3x = 3\times9.6\times10^{-4} = 2.88\times10^{-3}$ m

11. **[NYJC/Prelims 2014/P3/7]**

a. i. See definitions in lecture notes

ii. See definitions in lecture notes

b. i. 1. Waves from S_1 and S_2 arriving at M have a phase difference of 180° (or π rad).

OR Waves from S_1 and S_2 arriving at M have a path difference that is an odd integral of half wavelength.

2. Waves from S_1 and S_2 arriving at M have the same amplitude (or intensity).

OR Sources S_1 and S_2 have a ratio of amplitudes of 1.28.

ii. Path difference between waves from S_1 and S_2 is 28 cm.

Wavelength of the sources changes from 33 cm to 8.25 cm.

$$(2n+1)\pi = \frac{0.28}{\left(\frac{330}{f}\right)}(2\pi)$$

For f = 1.0 kHz: n = 0.34

For f = 4.0 kHz: n = 2.9

Minima occurs when wavelength of the sources are (56 cm,) 18.7 cm, 11.2 cm, (8.0 cm, 6.22 cm ...)

Since n can only be an integer, minima occurs when n = 1 and n = 2. Hence 2 minima observed.

iii. Amplitude of resultant wave at minima is 0.6 units and at maxima is 3.4 units

Intensity $\propto$ amplitude^2

$$\frac{\text{intensity when maxima is detected}}{\text{intensity when minima is detected}} = \left(\frac{3.4}{0.6}\right)^2 = 32$$

c. i.

$$a\sin\theta = n\lambda$$

$$\left(\frac{10^{-3}}{550}\right)\sin 90 = n(644\times10^{-9})$$

$$n = 2.8$$

Highest order that can be seen is the 2nd order. Hence total number of bright lines observed is 5.

ii. 1. Since θ is greater, λ is also greater.

2. When n is larger, $\Delta\theta$ is larger, thus the greatest separation occurs in the second order.

iii. 0th order (or central) maxima remains in the same position.

Diffraction pattern will rotate through 90°

iv. Grating is not normal to incident light **OR** Screen is not parallel to the grating

12. **[PJC/Prelims 2014/P2/3]**

a. A maximum is obtained at R when the waves in the two paths meet and superpose constructively (constructive interference) occurs. The path difference between the direct wave from T and R, and the reflected wave (reflected at ground) is equivalent to integer multiple of λ.

b. Minimum path difference = 1 λ = 2000 m

(Distance of path taken by reflected wave) – (2.4 × 10^4) = 1 λ

(Distance of path taken by reflected beam) – (2.4 × 10^4) = 2000 m

Distance of path taken by reflected beam = 26000 m

c. i. $f = \frac{c}{\lambda} = \frac{3\times10^8}{2.0\times10^3} = 1.5\times10^5$ Hz

ii. The resonator is configured to resonate at a particular frequency, allowing the antenna to amplify sinusoidal waves at that radio wave frequency and ignore other waves.

13. **[SAJC/Prelims 2014/P3/6 Part]**

c. i. $\lambda = \frac{ax}{D} \Rightarrow 590\times10^{-9} = \frac{(1.4\times10^{-3})x}{2.6} \Rightarrow x = 1.10\times10^{-3}$ m

ii. 180°

iii. A_{max} = 3.4, A_{min} = 0.6

$I = A^2$

Ratio = 3.4^2 / 0.6^2 = 32

d. i. $a\sin\theta = n\lambda \Rightarrow (650\times10^3)^{-1}\times\sin 90° = n\times590\times10^{-9} \Rightarrow n = 2.6$

Number of orders is 2.

ii. The maximum intensity produced through a diffraction grating is produced by constructive interference of light beams from a larger number of slits. The maximum intensity produced through a double-slit is produced by light beams from only two slits.

iii. The central fringe appears white since the path difference is zero for all colours here.

For higher orders, coloured spectrum are seen since the fringe separation for each colour depends on its wavelength according to $a\sin\theta = n\lambda$

14. **[SRJC/Prelims 2014/P2/5]**

a. A stationary wave is formed when 2 identical waves with same amplitude, frequency and wavelength (or speed) travelling in opposite directions towards each other and superpose.

b. i. 0

ii. 3

iii. A solid straight line across the centre of the wave (i.e. at the dotted line)

15. **[SRJC/Prelims 2014/P3/7]**

a. See definitions in lecture notes.

b. i. The two small slits allow diffraction of the light reaching it so that the light from both slits will spread and meet each other. Interference occurs in the region between the two slits and the screen where the diffracted waves meet.(inferred when they say meet each other)

Where the two waves arrive in phase, constructive interference takes place and a bright fringe is formed. Where the two waves arrive in antiphase, destructive interference takes place and a dark fringe is formed.

ii. $x = \lambda D/a \Rightarrow (10/4)\times10^{-3} = \lambda(2)/(0.5\times10^{-3}) \Rightarrow \lambda = 6.25\times10^{-7}$ m

iii. The spacing between the fringes will be reduced , but there is an increase in the intensity at the bright fringes only.

iv. Assuming original amplitude of both waves is A.

For maxima, the resultant amplitude is $2A$ and the original $I = 4kA^2$, where k is a constant,

When the intensity is now halved, $\frac{1}{2}(kA^2) = kA_{min}^{\ 2} \Rightarrow A_{min} = \frac{1}{\sqrt{2}}A$

Resultant amplitude of new minima $A_R = A - A_{min} = \left(1-\frac{1}{\sqrt{2}}\right)A$

Intensity of new minima = $\left[\left(1-\frac{1}{\sqrt{2}}\right)A\right]^2 = 0.085786A^2 = 0.0214(4A^2) = 0.0214I$

v. The red and blue light will overlap, resulting in a violet fringe, with a red tinge at the edges. The red tinge is due to the fact that the fringe separation, and hence fringe width of the red light is larger because of its larger wavelength.

c. i. Headlights are not coherent sources and therefore incapable of producing stable interference patterns.

ii. Since the distance between the speakers (2 m) is less than half a wavelength (2.5 m), the path difference is always less than half wavelength and hence, it is not possible to find a position where destructive interference will take place and hence a minima.

d. i. $a \sin\theta = n\lambda \Rightarrow 1/(2750 \times 10^2) \sin\theta = 2\,(700 \times 10^{-9}) \Rightarrow \theta = 22.6°$

$\tan\theta = 3/L \Rightarrow L = 7.19$ cm

ii. The separation distance of the ribs is much larger than the wavelength of the x-rays, hence there are no observable effects and hence the ribs do not act as a diffraction grating.

16. [TJC/Prelims 2014/P3/6]

a. See definitions in lecture notes.

b. Sources of waves must be coherent

Waves must have equal or similar amplitudes

waves must be polarized in the same plane or both unpolarized for transverse waves

waves must meet / waves must be of the same (transverse/longitudinal) type

c. The path difference from the two sources to the point must be equal to $n\lambda$ where n is an integer.

d. i. When the speaker is moved from **A** to **B**, the path difference varies leading to a varying vector addition (interference) of the signals the CRO ranging from fully constructive (maxima) to fully destructive (minima).

ii. In this case, the path difference is constant, hence there is no variation.

(Proof : Speaker at C, path difference = $M_1S - M_2S = M_1C - M_2C = M_1M_2$,

Speaker at D, path difference = $M_1S' - M_2S' = M_1D - M_2D = M_1M_2$)

e. i. As a second maxima is achieved when speaker moves by 10 cm in 1 s,

$\lambda = 10$ cm $= 0.10$ m

$v = f\lambda = 3000(0.10) = 300$ m s^{-1}

ii. Intensity of waves drop when they are further away from the source due to loss (attenuation) and the fact that the wave energy is spread over a larger area.

Since waves from S_1 and S_2 travel different distances to reach M, hence they don't have same amplitudes and hence do not cancel completely at minima points.

f. i. $a \sin\theta = n\lambda$

$(0.001/600) \sin\theta = 400 \times 10^{-9}$

$\theta = 13.9°$

$\tan\theta = \tan 13.9° = x/2.0$

$x = 0.495$ m

iii $a \sin\theta = n\lambda \Rightarrow 9.02 \times 10^{-7} \times \sin 90° = n \times 400 \times 10^{-9} \Rightarrow n = 2.26$

Maximum number of orders = 2

iii. new angle of deflection, $\phi = \tan^{-1}\left(\dfrac{2 \times 0.495}{2.0}\right) = 26.34°$

$a' \sin 26.34° = 400 \times 10^{-9}$

$a' = 9.02 \times 10^{-7}$ m

No. of lines per mm = $0.001 / (9.04 \times 10^{-7}) = 1110$

17. [TPJC/Prelims 2014/P2/3 Part]

b. i. Path length from each speaker to A is the same (or distance L_1A and L_2A same/equal).

Maximum at A, so waves must be in phase at A (or constructive interference).

Hence waves emitted from speakers are in phase (or zero phase difference).

ii. $\lambda = v / f = 340/6000 = 0.0567$ m

$x = \lambda D / a = 0.567$ m

Therefore, the distance moved from A towards P before encountering the minimum sound intensity $= 0.567 / 2 = 0.284$ m

18. [TPJC/Prelims 2014/P2/8 Part]

a. See definitions in lecture notes.

b. i. $x = \lambda D / a = (589 \times 10^{-9} \times 2.50) / 0.0008 = 0.00184$ m

ii. 1. The fringes have larger separation.

Because red light has longer wavelength

2. Fringe separation gets progressively larger towards the top.

Because D is getting larger.

3. The fringes will show both vertical and horizontal fringes arranged in a mesh-like pattern.

c. i. $a \sin\theta = n\lambda \Rightarrow a \sin 28° = 1 \times 430 \times 10^{-9} \Rightarrow a = 9.159 \times 10^{-7}$ m

number of lines per cm = $1 \times 10^{-2} / a = 1 \times 10^{-2} / 9.159 \times 10^{-7} = 1.09 \times 10^4$

ii. $a \sin\theta = n\lambda$

$(9.159 \times 10^{-7}) \sin(28 + \alpha) = 1 \times 680 \times 10^{-9}$

$\alpha = 19.9°$

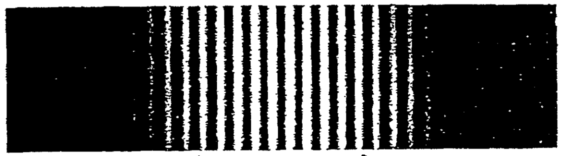
19. [VJC/Prelims 2014/P3/5 part]

b. i. See definitions in lecture notes.

When monochromatic light passes through the two closely spaced slits, the light waves from the two slits will superpose. Bright fringes are formed at points where the two waves meet in phase and interfere constructively.

Dark fringes are formed at points where the two waves meet out of phase and interfere destructively.

ii.



4.5 cm

Since photograph is full-scale, by measuring using a ruler as shown:

13 fringe spacings = 4.5 cm

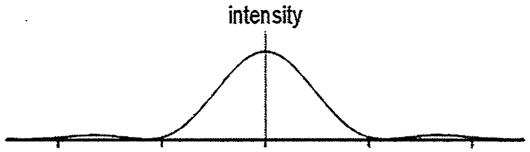
Hence fringe spacing $x = 4.5/13 = 0.35$ cm $= 3.5 \times 10^{-3}$ m iii.

$$x = \frac{\lambda D}{a}$$

$$\lambda = \frac{x \times a}{D} = \frac{3.5\times10^{-3}\times0.30\times10^{-3}}{1.5} = 7.00\times10^{-7}\ \text{m}$$

iv. This is because the intensity of the double-slit interference pattern is restricted by that of the single-slit diffraction pattern.

v.



20. **[YJC/Prelims 2014/P2/3]**

a. See definitions in lecture notes.

b. i. Light of single wavelength

ii. Gradient = $2 \times 10^{-3} / 1 = 2 \times 10^{-3}$

iii. $x = \lambda D / a$

grad = λ / a

$a = \lambda$ / grad = $(620 \times 10^{-9})/(2 \times 10^{-3}) = 3.1 \times 10^{-4}$ m

iv. Steeper gradient

21. **[YJC/Prelims 2014/P3/4]**

a. i. $a \sin \theta = n\lambda \Rightarrow a \sin 28° = 1 \times 430 \times 10^{-9} \Rightarrow a = 9.159 \times 10^{-7}$ m

number of slits per metre = $1 / a = 1 \times 10^{-2} / 9.159 \times 10^{-7} = 1.09 \times 10^{6}$

ii. $a \sin \theta = n\lambda$

$(9.159 \times 10^{-7}) \sin (28 + \alpha) = 1 \times 680 \times 10^{-9}$

$\alpha = 19.9°$

b. The central fringe will be white as all wavelengths of the white beam will arrive and merge together to form white light

c. Certain wavelengths emerge at the same angular separation from the central fringe.

d. $a \sin 90° = n (600 \times 10^{-9})$

$n = 1.53$

Maximum order = 1

Total bright lines = 3